

# IIT MATHEMATICS

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## FUNCTION

## WORKBOOK

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# RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



**AIR-1**  
**CHITRAANG MURDIA**  
Gen. Category - 2014



**AIR-3**  
**GOVIND LAHOTI**  
Gen. Category - 2014

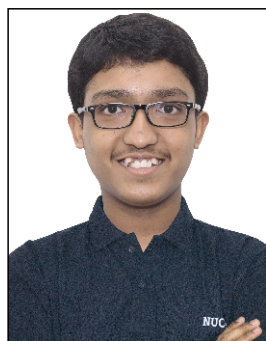


**AIR-6**  
**NISHIT AGARWAL**  
Gen. Category - 2012

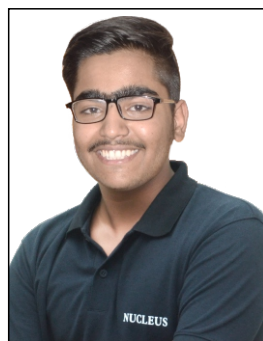


**AIR-8**  
**AMEY GUPTA**  
Gen. Category - 2014

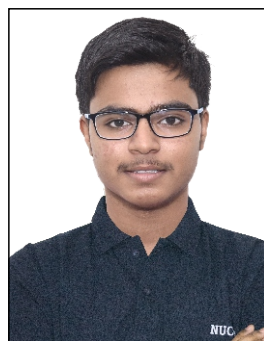
## JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



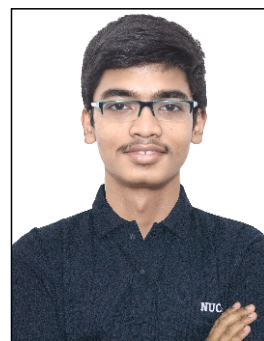
**100 Percentile**  
**(Physics)**  
**DAKSH KHANDLWAL**  
2020



**100 Percentile**  
**(Maths & Physics)**  
**VAIBHAV SAHA**  
2020



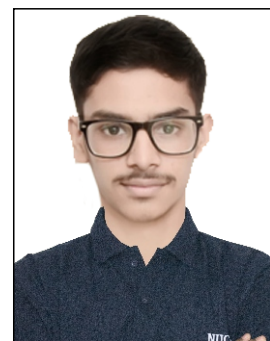
**100 Percentile**  
**(Physics)**  
**ANISH MOHAN**  
2020



**100 Percentile**  
**(Maths)**  
**ARCHIT PATNAIK**  
2020

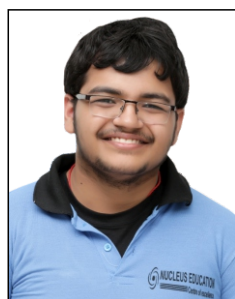


**100 Percentile**  
**(Maths)**  
**SWAPNIL YASASVI**  
2020

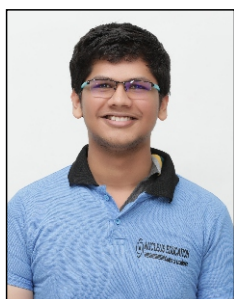


**100 Percentile**  
**(Maths)**  
**PARSHANT ARORA**  
2020

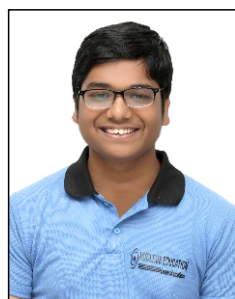
## JEE ADVANCED RESULTS OF NUCLEUS EDUCATION



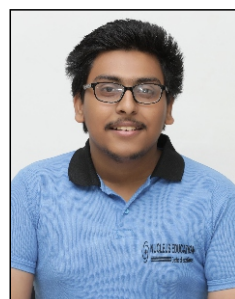
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**LAKSHAY SHARMA**  
Gen. Category - 2017



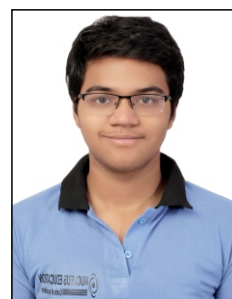
**AIR-12**  
**YATEESH AGRAWAL**  
Gen. Category - 2017



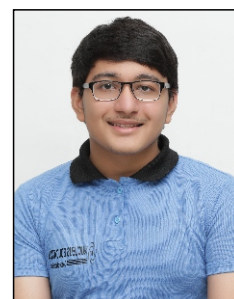
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**ABHEY GOYAL**  
Gen. Category - 2017



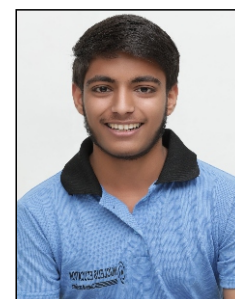
**AIR-24**  
**TUSHAR GAUTAM**  
Gen. Category - 2017



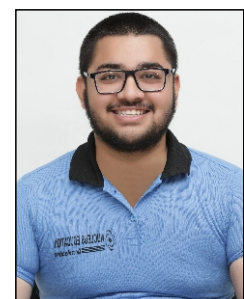
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**PIYUSH TIBAREWAL**  
Gen. Category - 2017



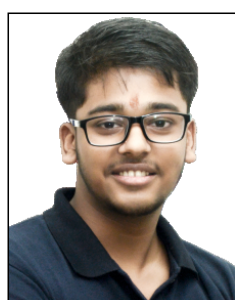
**AIR-42**  
**SATVIK**  
Gen. Category - 2017



**AIR-66**  
**MAYANK DUBEY**  
Gen. Category - 2017



**AIR-98**  
**HRITHIK**  
Gen. Category - 2017



**AIR-20**  
**SHASHANK AGRAWAL**  
Gen. Category - 2018



**AIR-27**  
**RAAGHAV RAAJ**  
Gen. Category - 2018



**AIR-32**  
**SHREYA PATHAK**  
Gen. Category - 2018



**AIR-61**  
**SIDDHANT CHOUDHARY**  
Gen. Category - 2018



**AIR-67**  
**ANISWAR S KRISHNAN**  
Gen. Category - 2018  
DLP



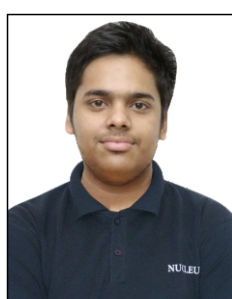
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**AAYUSH KADAM**  
Gen. Category - 2018



**AIR-61**  
**SARTHAK BEHERA**  
Gen. Category - 2018  
\*SDCCP



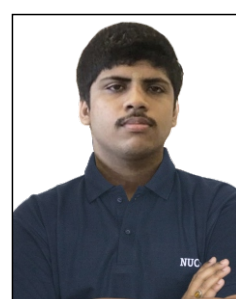
**AIR-91**  
**ANDREWS G. VARGHESE**  
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\*SDCCP



**AIR-2**  
**HIMANSHU GAURAV SINGH**  
Gen. Category - 2019  
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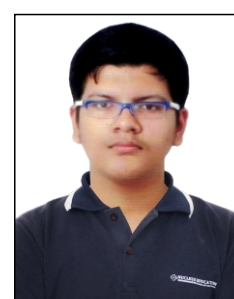
**AIR-19**  
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Gen. Category - 2019



**AIR-33**  
**S. PRAJEETH**  
Gen. Category - 2019  
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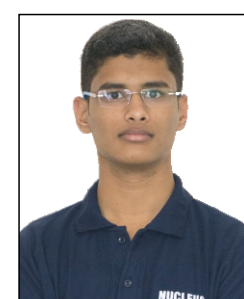
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**SOHAM MISTRI**  
Gen. Category - 2019  
DLP



**AIR-51**  
**SAYANTAN DHAR**  
Gen. Category - 2019  
DLP



**AIR-53**  
**GAURAV KRISHAN GUPTA**  
Gen. Category - 2019  
DLP



**AIR-86**  
**SATVIK JAIN**  
Gen. Category - 2019

## FUNCTION

### 1. CARTESIAN PRODUCT OF TWO SETS

Given two non-empty sets A and B. The cartesian product  $A \times B$  is the set of all ordered pairs of the form (a, b) where the first entry comes from set A & second comes from set B.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

e.g.  $A = \{1, 2, 3\}$   $B = \{p, q\}$

$$A \times B = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\}$$

**Note :**

- (i) If either A or B is the null set, then  $A \times B$  will also be empty set, i.e.  $A \times B = \phi$
- (ii) If  $n(A) = p$ ,  $n(B) = q$ , then  $n(A \times B) = pq$ , where  $n(X)$  denotes the number of elements in set X.
- (iii) A **Relation** R from set A to B is any subset of  $A \times B$ . If  $A \times B$  &  $(a, b) \in R$  then b is image of a under R and a is preimage of b under R.

**Note :** If  $n(A) = m$ ,  $n(B) = n$ , then number of relations defined from set A to B are  $2^{mn} - 1$ .

### 2. FUNCTION :

A relation R from set A to set B is called a function if each element of A is uniquely associated with some element of B. It is denoted by the symbol :

$$f : A \rightarrow B \text{ or } A \xrightarrow{f} B$$

which reads 'f' is a function from A to B 'or' f maps A to B,

If an element  $a \in A$  is associated with an element  $b \in B$ , then b is called 'the f image of a' or 'image of a under f' or 'the value of the function f at a'. Also a is called the 'pre-image of b' or 'argument of b under the function f'.

We write it as

$$b = f(a) \text{ or } f : a \rightarrow b \text{ or } f : (a, b)$$

Thus a function 'f' from set A to set B is subset of  $A \times B$  in which each a belonging to A appears in one and only one ordered pair belonging to f.

### Representation of Function :

**(a) Ordered pair :** Every function from

$A \rightarrow B$  satisfies the following conditions :

- (i)  $f \subset A \times B$
- (ii)  $\forall a \in A$  there exist  $b \in B$  and
- (iii)  $(a, b) \in f \text{ \& } (a, c) \in f \Rightarrow b = c$

**(b) Formula based**

**(uniformly/ nonuniformly) :**

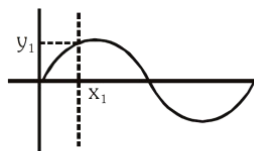
(i)  $f : \mathbb{R} \rightarrow \mathbb{R}, y = f(x) = 4x, f(x) = x^2$   
(uniformly defined)

(ii)  $f(x) = \begin{cases} x+1 & -1 \leq x < 4 \\ -x & 4 \leq x < 7 \end{cases}$   
(non-uniformly defined)

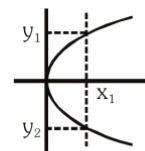
(iii)  $f(x) = \begin{cases} x^2 & x \geq 0 \\ -x-1 & x < 0 \end{cases}$   
(non-uniformly defined)

**(c) Graphical representation :**

If a vertical line cuts a given graph at more than one point then it can not be the graph of a function.



Graph (1)



Graph (2)

Graph(1) represent a function but graph(2) does not represent a function.

Every function is a relation but every relation is not necessarily a function.

### 3. DOMAIN, CO-DOMAIN & RANGE OF A FUNCTION :

Let  $f : A \rightarrow B$ , then the set  $A$  is known as the domain of  $f$  & the set  $B$  is known as co-domain of  $f$ . The set of  $f$  images of all the elements of  $A$  is known as the range of  $f$ .

Domain of  $f = \{a \mid a \in A, (a, f(a)) \in f\}$

Range of  $f = \{f(a) \mid a \in A, f(a) \in B\}$

- (a) If only the rule of function is given then the domain of the function is the set of those real numbers, where function is defined.
- (b) For a continuous function, the interval from minimum to maximum value of a function gives the range
- (c) It should be noted that range is a subset of co-domain.

#### Note :

- (i) The complete set of all positive real numbers is denoted by  $R^+$ .
- (ii) The complete set of all negative real numbers is denoted by  $R^-$ .
- (iii) The complete set of all real numbers other than zero is denoted by  $R_0$ .
- (iv) The complete set of all integers is denoted by  $Z$ .

**Illustration 1 :** Find the domain of following functions :

$$(i) \quad y = \sqrt{5-2x} \quad (ii) \quad y = \frac{1}{\sqrt{x-|x|}}$$

**Solution :** (i)  $5 - 2x \geq 0 \Rightarrow x \leq \frac{5}{2}$

$\therefore$  Domain is  $(-\infty, 5/2]$

- (ii)  $x - |x| > 0 \Rightarrow |x| < x \Rightarrow x$  cannot take any real values  $\therefore$  Domain is  $\phi$

**Illustration 2 :** Find the range of following functions :

$$(i) \quad f(x) = \log_{\sqrt{2}}((x-1)^2 + 4)$$

$$(ii) \quad f(x) = 3 - \cos x$$

**Solution :**

- (i)  $f(x) = \log_{\sqrt{2}}((x-1)^2 + 4)$   
 $4 \leq (x-1)^2 + 4 < \infty$   
 $\Rightarrow \log_{\sqrt{2}} 4 \leq \log_{\sqrt{2}}((x-1)^2 + 4) < \infty$   
 $\Rightarrow 4 \leq \log_{\sqrt{2}}((x-1)^2 + 4) < \infty$   
 $\therefore$  Range of  $f(x) = [4, \infty)$
- (ii)  $f(x) = 3 - \cos x$   
 $-1 \leq \cos x \leq 1$   
 $2 \leq 3 - \cos x \leq 4$   
 $\therefore$  Range of  $f(x) = [2, 4]$

#### Do yourself -1

(i) Find the domain of following functions :

$$(a) \quad y = 1 - \log_{10} x \quad (b) \quad y = \frac{1}{\sqrt{x^2 - 4x}}$$

(ii) Find the range of the following function :

- (a)  $\log_4 \left| x + \frac{1}{x} \right|$
- (b)  $f(x) = \sin(3x^2 + 1)$
- (c)  $f(x) = 2\sin\left(2x + \frac{\pi}{4}\right)$
- (d)  $f(x) = \cos\left(2x + \frac{\pi}{4}\right)$

### 4. IMPORTANT TYPES OF FUNCTIONS :

(a) **Polynomial Function :**

If a function  $f$  is defined by

$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$  where  $n$  is a non negative integer and  $a_0, a_1, a_2, \dots, a_n$  are real numbers and  $a_0 \neq 0$ , then  $f$  is called a polynomial function of degree  $n$ . If  $n$  is odd, then polynomial is of odd degree, if  $n$  is even, then polynomial is of even degree.

**Note :**

- (i) Range of odd degree polynomial is always  $\mathbb{R}$ .  
 (ii) Range of even degree polynomial is never  $\mathbb{R}$ .  
 (iii) A polynomial of degree one with no constant term is called an odd linear function. i.e.  
 $f(x) = ax$ ,  $a \neq 0$   
 (iv)  $f(x) = ax + b$ ,  $a \neq 0$  is a linear polynomial  
 (v)  $f(x) = c$  is non linear polynomial (its degree is zero)  
 (vi)  $f(x) = 0$  is a polynomial but its degree is **not** defined  
 (vii) There are two polynomial functions, satisfying the relation ;

$$f(x) \cdot f(1/x) = f(x) + f(1/x). \text{ They are :}$$

(a)  $f(x) = x^n + 1$  &

(b)  $f(x) = 1 - x^n$ , where  $n$  is a positive integer.

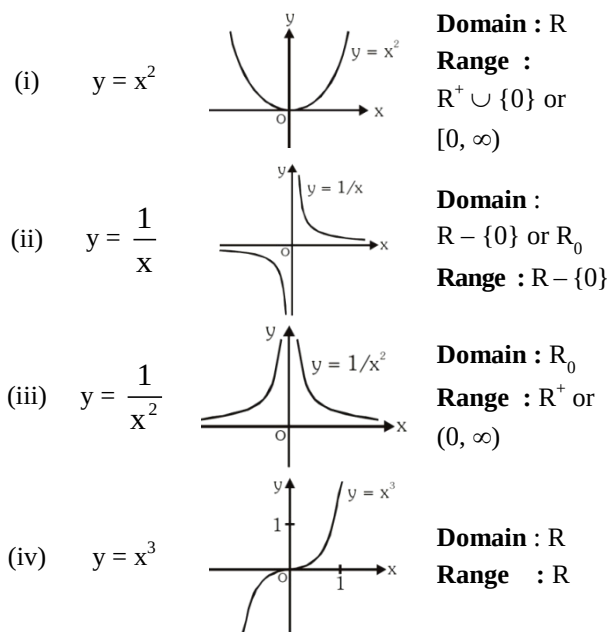
- (b) **Algebraic Function :** A function ' $f$ ' is called an algebraic function if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking radicals) within polynomials.

**Example :**  $f(x) = \sqrt{x^2 - 1}$ ;

$$g(x) = \frac{x^4 - 16x^2}{x + \sqrt{x}} + (x-2)\sqrt[3]{x+1}$$

$y$  is an algebraic function of  $x$ , if it is a function that satisfies an algebraic equation of the form  $P_0(x)y^n + P_1(x)y^{n-1} + \dots + P_{n-1}(x)y + P_n(x) = 0$ . Where  $n$  is a positive integer and  $P_0(x), P_1(x), \dots$  are Polynomials in  $x$ . e.g.  $x^3 + y^3 - 3xy = 0$ .

- Note :** (i) All polynomial functions are Algebraic but not the converse.  
 (ii) A function that is not algebraic is called **Transcendental Function**.

**Basic algebraic function :****(c) Rational function :**

A rational function is a function of the form

$$y = f(x) = \frac{g(x)}{h(x)}, \text{ where } g(x) \text{ \& } h(x) \text{ are polynomials \& } h(x) \neq 0, \text{ Domain : } \mathbb{R} - \{x \mid h(x) = 0\}$$

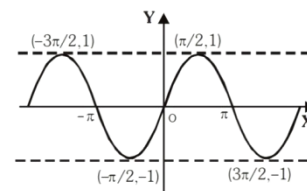
Any rational function is automatically an algebraic function.

**(d) Trigonometric function :****(i) Sine function**

$$f(x) = \sin x$$

**Domain :**  $\mathbb{R}$

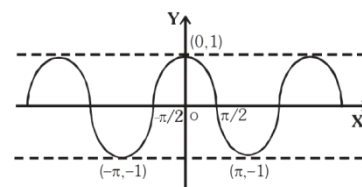
**Range :**  $[-1, 1]$ , period  $2\pi$

**(ii) Cosine function**

$$f(x) = \cos x$$

**Domain :**  $\mathbb{R}$

**Range :**  $[-1, 1]$ , period  $2\pi$



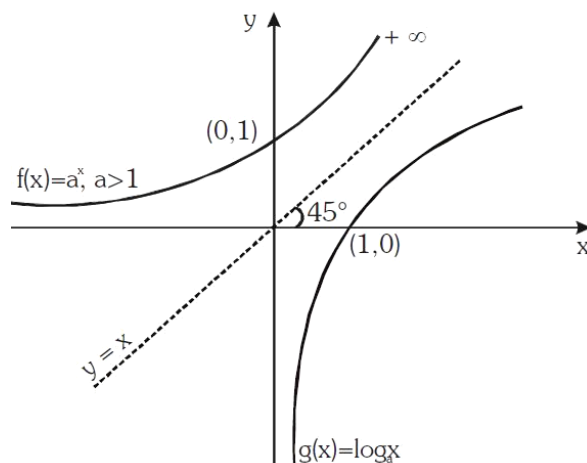
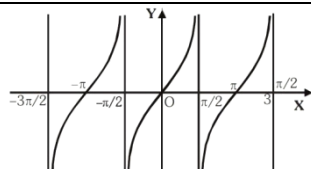
(iii) **Tangent function**

$$f(x) = \tan x$$

**Domain :**

$$\mathbb{R} - \left\{ x \mid x = \frac{(2n+1)\pi}{2}, n \in \mathbb{I} \right\}$$

**Range :**  $\mathbb{R}$ , period  $\pi$

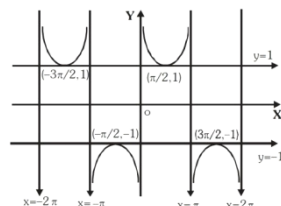


(iv) **Cosecant function**

$$f(x) = \operatorname{cosec} x$$

**Domain :**  $\mathbb{R} - \{x \mid x = n\pi, n \in \mathbb{I}\}$

**Range :**  $\mathbb{R} - (-1, 1)$ , period  $2\pi$

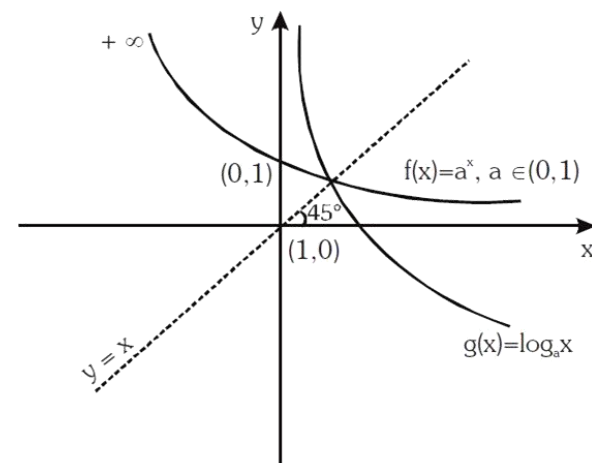
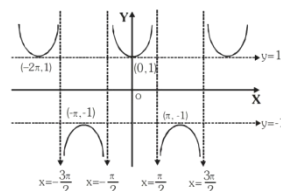


(v) **Secant function**

$$f(x) = \sec x$$

**Domain :**  $\mathbb{R} - \{x \mid x = (2n+1)\pi/2, n \in \mathbb{I}\}$

**Range :**  $\mathbb{R} - (-1, 1)$ , period  $2\pi$

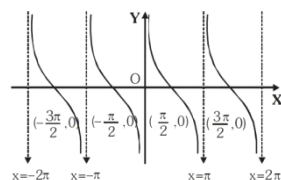


(vi) **Cotangent function**

$$f(x) = \cot x$$

**Domain :**  $\mathbb{R} - \{x \mid x = n\pi, n \in \mathbb{I}\}$

**Range :**  $\mathbb{R}$ , period  $\pi$



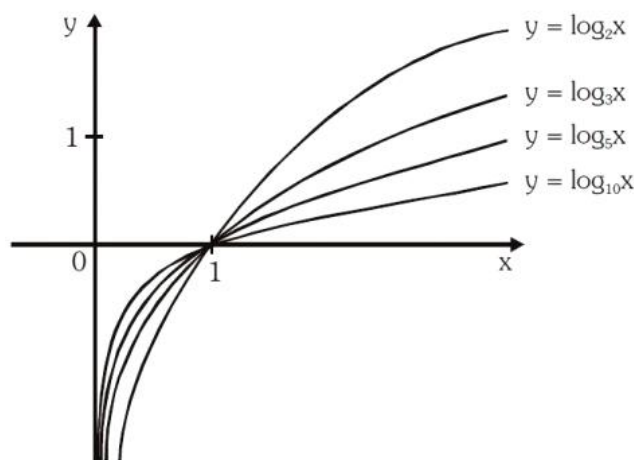
(e) **Exponential and Logarithmic Function :**

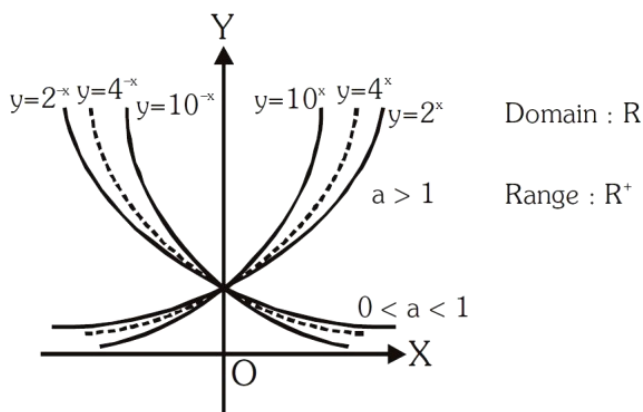
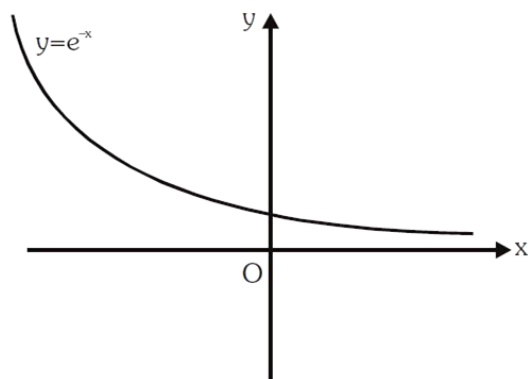
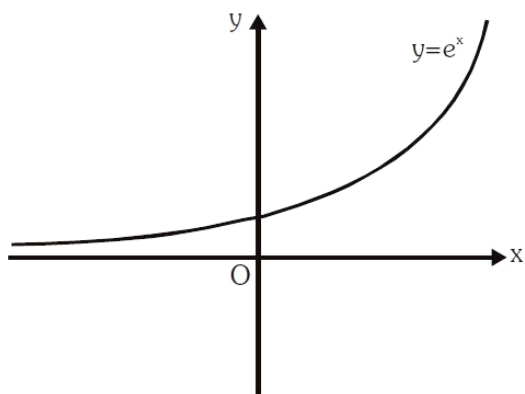
A function  $f(x) = a^x$  ( $a > 0$ ),  $a \neq 1$ ,  $x \in \mathbb{R}$  is called an exponential function. The inverse of the exponential function is called the logarithmic function, i.e.  $g(x) = \log_a x$ .

Note that  $f(x)$  &  $g(x)$  are inverse of each other & their graphs are as shown. (If functions are mirror image of each other about the line  $y = x$ )

**Domain** of  $a^x$  is  $\mathbb{R}$  **Range**  $\mathbb{R}^+$

**Domain** of  $\log_a x$  is  $\mathbb{R}^+$  **Range**  $\mathbb{R}$





**Note-1 :**  $f(x) = a^{1/x}$ ,  $a > 0$

**Domain :**  $\mathbb{R} - \{0\}$

**Range :**  $\mathbb{R}^+ - \{1\}$

**Note-2 :**  $f(x) = \log_a x = \frac{1}{\log_a x}$

**Domain :**  $\mathbb{R}^+ - \{1\}$

**Range :**  $\mathbb{R} - \{0\}$   
( $a > 0$ ) ( $a \neq 1$ )

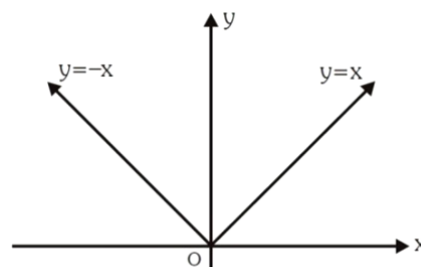
### (f) Absolute Value Function :

A function  $y = f(x) = |x|$  is called the absolute value function or Modulus function. It is defined as :

$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

For  $f(x) = |x|$ , domain is  $\mathbb{R}$  and range is  $[0, \infty)$

For  $f(x) = \frac{1}{|x|}$ , domain is  $\mathbb{R} - \{0\}$  and range is  $\mathbb{R}^+$ .



### (g) Signum Function :

A function  $y = f(x) = \text{Sgn}(x)$  is defined as follows :

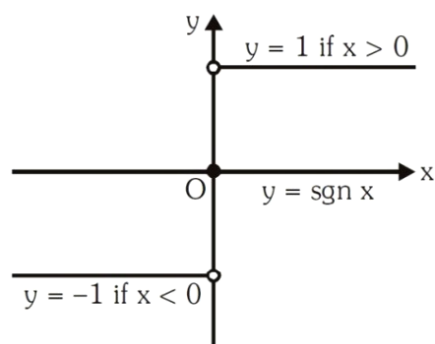
$$y = f(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$

It is also written as  $\text{Sgn } x = |x|/x$ ;  $x \neq 0$

**Note :**  $f(x) = (\text{sgn}(x))x \Rightarrow f(x) = |x|$

**Domain :**  $\mathbb{R}$

**Range :**  $\{-1, 0, 1\}$

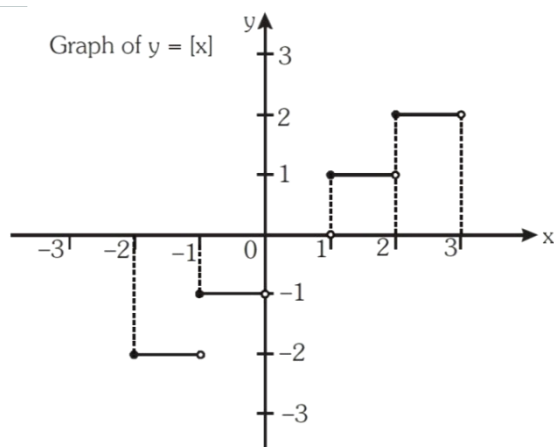


**(h) Greatest integer or step up function :**

The function  $y = f(x) = [x]$  is called the greatest integer function where  $[x]$  denotes the greatest integer less than or equal to  $x$ .

Note that for :

| $x$        | $[x]$ |
|------------|-------|
| $[-2, -1)$ | -2    |
| $[-1, 0)$  | -1    |
| $[0, 1)$   | 0     |
| $[1, 2)$   | 1     |



**Domain :**  $\mathbb{R}$

**Range :**  $\mathbb{I}$

**Properties of greatest integer function :**

(i)  $[x] \leq x < [x] + 1$  and  $x - 1 < [x] \leq x$ ,

$0 \leq x - [x] < 1$

(ii)  $[x + m] = [x] + m$ , if  $m$  is an integer.

(iii)  $[x] + [-x] = \begin{cases} 0, & x \in \mathbb{I} \\ -1, & x \notin \mathbb{I} \end{cases}$

**Illustration 3 :** If  $y = 2[x] + 3$  &  $y = 3[x - 2] + 5$ , then find  $[x + y]$  where  $[.]$  denotes greatest integer function.

**Solution :**  $y = 3[x - 2] + 5 = 3[x] - 1$

so  $3[x] - 1 = 2[x] + 3$

$[x] = 4 \Rightarrow 4 \leq x < 5$  then  $y = 11$

so  $x + y$  will lie in the interval  $[15, 16)$

so  $[x + y] = 15$

**Illustration 4 :** Find the value of

$$\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{2946}{1000}\right]$$

where  $[.]$  greatest integer function ?

**Solution :**

$$\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{499}{1000}\right] + \left[\frac{1}{2} + \frac{500}{1000}\right]$$

$$+ \dots + \left[\frac{1}{2} + \frac{1499}{1000}\right] + \left[\frac{1}{2} + \frac{1500}{1000}\right]$$

$$+ \dots + \left[\frac{1}{2} + \frac{2499}{1000}\right] + \left[\frac{1}{2} + \frac{2500}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{2946}{1000}\right]$$

$$= 0 + 1 \times 1000 + 2 \times 1000 + 3 \times 447$$

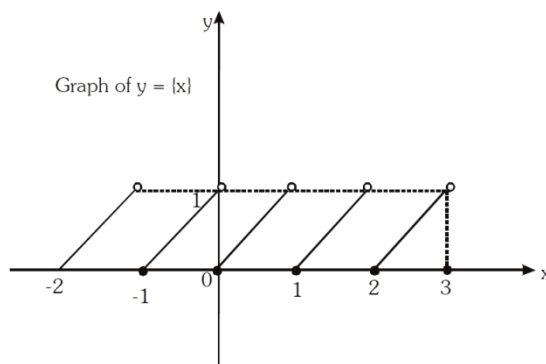
$$= 3000 + 1341 = 4341$$

**Ans.**

**(i) Fractional part function :**

It is defined as :  $g(x) = \{x\} = x - [x]$  e.g. the fractional part of the number 2.1 is  $2.1 - 2 = 0.1$  and the fractional part of  $-3.7$  is  $0.3$ . The period of this function is 1 and graph of this function is as shown.

| $x$        | $\{x\}$ |
|------------|---------|
| $[-2, -1)$ | $x + 2$ |
| $[-1, 0)$  | $x + 1$ |
| $[0, 1)$   | $x$     |
| $[1, 2)$   | $x - 1$ |



**Domain :**  $\mathbb{R}$

**Range :**  $[0, 1)$

**Properties of fractional part function :**

- (i)  $0 \leq \{x\} < 1$
- (ii)  $\{[x]\} = \{ \{x\} \} = 0$
- (iii)  $\{ \{x\} \} = \{x\}$
- (iv)  $\{x + m\} = \{x\}, m \in \mathbb{I}$
- (v)  $\{x\} + \{-x\} = \begin{cases} 1, & x \notin \mathbb{I} \\ 0, & x \in \mathbb{I} \end{cases}$

**Illustration 5 :** Solve the equation

$|2x - 1| = 3[x] + 2\{x\}$  where  $[.]$  denotes greatest integer and  $\{.\}$  denotes fractional part function.

**Solution :** We are given that,

$$|2x - 1| = 3[x] + 2\{x\}$$

Let,  $2x - 1 \leq 0$  i.e.  $x \leq \frac{1}{2}$ . The given equation

yields.

$$1 - 2x = 3[x] + 2\{x\}$$

$$\Rightarrow 1 - 2[x] - 2\{x\} = 3[x] + 2\{x\}$$

$$\Rightarrow 1 - 5[x] = 4\{x\}$$

$$\Rightarrow \{x\} = \frac{1 - 5[x]}{4}$$

$$\Rightarrow 0 \leq \frac{1 - 5[x]}{4} < 1$$

$$\Rightarrow 0 \leq 1 - 5[x] < 4$$

$$\Rightarrow -\frac{3}{5} < [x] \leq \frac{1}{5}$$

Now,  $[x] = 0$  as zero is the only integer lying

between  $-\frac{3}{5}$  and  $\frac{1}{5}$

$$\Rightarrow \{x\} = \frac{1}{4}$$

$$\Rightarrow x = \frac{1}{4} \text{ which is less than } \frac{1}{2},$$

Hence  $\frac{1}{4}$  is one solution.

$$\text{Now, let } 2x - 1 > 0 \text{ i.e. } x > \frac{1}{2}$$

$$\Rightarrow 2x - 1 = 3[x] + 2\{x\}$$

$$\Rightarrow 2[x] + 2\{x\} - 1 = 3[x] + 2\{x\}$$

$$\Rightarrow [x] = -1$$

$$\Rightarrow -1 \leq x < 0 \text{ which is not a solution as } x > \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{4} \text{ is the only solution.}$$

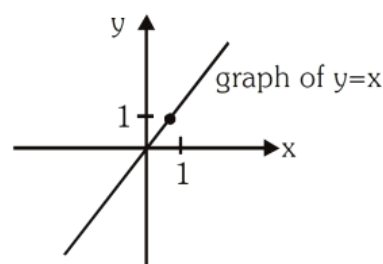
**(j) Identity function :**

The function  $f : A \rightarrow A$  defined by  $f(x) = x$

$\forall x \in A$  is called the identity of  $A$  and is

denoted by  $I_A$ . It is easy to observe that

identity function defined on  $\mathbb{R}$  is a bijection.



**(k) Constant function :**

A function  $f : A \rightarrow B$  is said to be a

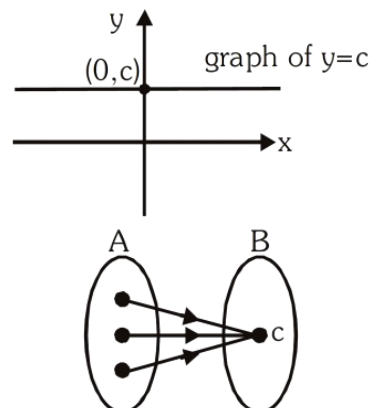
constant function if every element of  $A$  has

the same  $f$  image in  $B$ . Thus  $f : A \rightarrow B ; f$

$(x) = c, \forall x \in A, c \in B$  is a constant

function. Note that the range of a constant

function is a singleton.



**Do yourself -2**

- (i) Let  $\{x\}$  &  $[x]$  denotes the fraction and integral part of a real number  $x$  respectively, then match the column.

**Column-I**

**Column-II**

- |                            |                                            |
|----------------------------|--------------------------------------------|
| (A) $[x^2] \geq 4$         | (p) $x \in [2, 4]$                         |
| (B) $[x]^2 - 5[x] + 6 = 0$ | (q) $x \in (-\infty, -2] \cup [2, \infty)$ |
| (C) $x = \{x\}$            | (r) $x \in (-\infty, -5)$                  |
| (D) $[x] < -5$             | (s) $x \in \{-2\}$                         |
|                            | (t) $x \in [0, 1)$                         |

**5. ALGEBRAIC OPERATIONS ON FUNCTIONS**

If  $f$  &  $g$  are real valued functions of  $x$  with domain set  $A$ ,  $B$  respectively,

$f + g$ ,  $f - g$ ,  $(f \cdot g)$  &  $(f/g)$  as follows :

(a)  $(f \pm g)(x) = f(x) \pm g(x)$

domain in each case is  $A \cap B$

(b)  $(f \cdot g)(x) = f(x) \cdot g(x)$

domain is  $A \cap B$

(c)  $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

domain  $A \cap B - \{x \mid g(x) = 0\}$

**Illustration 6 :** Find the domain of the following function :

(i)  $y = \log_{(x-4)}(x^2 - 11x + 24)$

(ii)  $f(x) = \log_2 \left( -\log_{1/2} \left( 1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$

**Solution :** (i)  $y = \log_{(x-4)}(x^2 - 11x + 24)$

Here 'y' would assume real value if,

$x - 4 > 0$  and  $\neq 1$ ,  $x^2 - 11x + 24 > 0$

$\Rightarrow x > 4$  and  $\neq 5$ ,  $(x - 3)(x - 8) > 0$

$\Rightarrow x > 4$  and  $\neq 5$ ,  $x < 3$  or  $x > 8$

$\Rightarrow x > 8$

$\Rightarrow \text{Domain } (y) = (8, \infty)$

(ii) We have  $f(x) = \log_2 \left( -\log_{1/2} \left( 1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$

$f(x)$  is defined if  $-\log_{1/2} \left( 1 + \frac{1}{\sqrt[4]{x}} \right) - 1 > 0$

or if  $\log_{1/2} \left( 1 + \frac{1}{\sqrt[4]{x}} \right) < -1$

or if  $\left( 1 + \frac{1}{\sqrt[4]{x}} \right) > (1/2)^{-1}$

or if  $1 + \frac{1}{\sqrt[4]{x}} > 2$

or if  $\frac{1}{\sqrt[4]{x}} > 1$

or if  $x^{1/4} < 1$

or if  $0 < x < 1$

$\therefore D(f) = (0, 1)$

**Illustration 7 :** Find the domain

$f(x) = \frac{1}{\sqrt{[|x| - 5] - 11}}$  where  $[.]$  denotes greatest integer function.

integer function.

**Solution :**  $[|x| - 5] > 11$

so  $[|x| - 5] > 11$  or  $[|x| - 5] < -11$

$[|x|] > 16$   $[|x|] < -6$

$|x| \geq 17$  or  $|x| < -6$  (Not possible)

$\Rightarrow x \leq -17$  or  $x \geq 17$

so  $x \in (-\infty, -17] \cup [17, \infty)$

**Illustration 8:** Find the range of following functions

(i)  $f(x) = \frac{1}{8 - 3\sin x}$

(ii)  $f(x) = \log_{\sqrt{2}} (2 - \log_2 (16\sin^2 x + 1))$

**Solution :** (i)  $f(x) = \frac{1}{8 - 3\sin x}$

$-1 \leq 3 \sin x \leq 1$

$\therefore \text{Range of } f = \left[ \frac{1}{11}, \frac{1}{5} \right]$

- (ii)  $f(x) = \log_{\sqrt{2}}(2 - \log_2(16\sin^2 x + 1))$   
 $1 \leq 16\sin^2 x + 1 \leq 17$   
 $\therefore 0 \leq \log_2(16\sin^2 x + 1) \leq \log_2 17$   
 $\therefore 2 - \log_2 17 \leq 2 - \log_2(16\sin^2 x + 1) \leq 2$   
 Now consider  $0 < 2 - \log_2(16\sin^2 x + 1) \leq 2$   
 $\therefore -\infty < \log_{\sqrt{2}}[2 - \log_2(16\sin^2 x + 1)] \leq \log_{\sqrt{2}} 2 = 2$   
 $\therefore$  the range is  $(-\infty, 2]$

**Illustration 9 :** Find the range of  $f(x) = \frac{x - [x]}{1 + x - [x]}$ ,

where  $[.]$  denotes greatest function.

**Solution :**  $y = \frac{x - [x]}{1 + x - [x]} = \frac{\{x\}}{1 + \{x\}}$   
 $\therefore \frac{1}{y} = \frac{1}{\{x\}} + 1 \Rightarrow \frac{1}{\{x\}} = \frac{1-y}{y} \Rightarrow \{x\} = \frac{y}{1-y}$   
 $0 \leq \{x\} < 1 \Rightarrow 0 \leq \frac{y}{1-y} < 1$

Range =  $[0, 1/2)$

### Do yourself -3

(i) Find domain of following functions :

(a)  $f(x) = \sin(\sqrt{1-x^2}) + \sqrt{x+2} + \frac{1}{\log_{10}(x+1)}$

(b)  $f(x) = \sqrt{\frac{(2x+1)}{x^3 - 3x^2 + 2x}}$

(ii) Find range of following functions :

(a)  $f(x) = \log_2(\log_{1/2}(x^2 + 4x + 4))$

(b)  $f(x) = \frac{1}{2 - \cos 3x}$

### 6. EQUAL OR IDENTICAL FUNCTION :

Two function  $f$  &  $g$  are said to be equal if :

- (a) The domain of  $f$  = the domain of  $g$   
 (b) The co-domain of  $f$  = co-domain of  $g$  and  
 (c)  $f(x) = g(x)$ , for every  $x$  belonging to their common domain (i.e. should have the same graph)

**Illustration 10 :** The function

$$f(x) = \log(x-1) - \log(x-2) \text{ and } g(x) = \log\left(\frac{x-1}{x-2}\right)$$

are identical when  $x$  lies in the interval

- (A)  $[1, 2]$  (B)  $[2, \infty)$   
 (C)  $(2, \infty)$  (D)  $(-\infty, \infty)$

**Solution :** Since  $f(x) = \log(x-1) - \log(x-2)$ .

Domain of  $f(x)$  is  $x > 2$  or  $x \in (2, \infty) \dots(i)$

$$g(x) = \log\left(\frac{x-1}{x-2}\right) \text{ is defined if } \frac{x-1}{x-2} > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (2, \infty) \dots(ii)$$

From (i) and (ii),  $x \in (2, \infty)$  **Ans. (C)**

### Do yourself -4

(i) Are the following functions identical ?

(a)  $f(x) = \frac{x}{x^2}$  &  $\phi(x) = \frac{x^2}{x}$

(b)  $f(x) = x$  &  $\phi(x) = \sqrt{x^2}$

(c)  $f(x) = \log_{10} x^2$  &  $\phi(x) = 2\log_{10}|x|$

### 7. HOMOGENEOUS FUNCTIONS :

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables.

For examples  $5x^2 + 3y^2 - xy$  is homogenous in  $x$  &  $y$ . Symbolically if,  $f(tx, ty) = t^n f(x, y)$  then  $f(x, y)$  is homogeneous function of degree  $n$ .

**Illustration 11 :** Which of the following function is not homogeneous ?

(A)  $x^3 + 8x^2y + 7y^3$  (B)  $y^2 + x^2 + 5xy$

(C)  $\frac{xy}{x^2 + y^2}$  (D)  $\frac{2x - y + 1}{2y - x + 1}$

**Solution :** It is clear that (D) does not have the same degree in each term. **Ans. (D)**

## 8. BOUNDED FUNCTION :

A function is said to be bounded if there exists a finite  $M$  such that  $|f(x)| \leq M, \forall x \in D_f$ .

## 9. IMPLICIT & EXPLICIT FUNCTION :

A function defined by an equation not solved for the dependent variable is called an **implicit function**. e.g. the equations  $x^3 + y^3 = 1$  &  $x^y = y^x$ , defines  $y$  as an implicit function. If  $y$  has been expressed in terms of  $x$  alone then it is called an **Explicit function**.

**Illustration 12 :** Which of the following function is implicit function ?

- (A)  $y = \frac{x^2 + e^x + 5}{\sqrt{1 - \cos^{-1} x}}$  (B)  $y = x^2$
- (C)  $xy - \sin(x + y) = 0$  (D)  $y = \frac{x^2 \log x}{\sin x}$

**Solution :** It is clear that in (C)  $y$  is not clearly expressed in  $x$ . **Ans. (C)**

### Do yourself -5

(i) Find the boundness of the function

$$f(x) = \frac{x^2}{x^4 + 1}$$

(ii) Which of the following function is implicit function ?

- (A)  $xy - \cos(x + y) = 0$
- (B)  $y = x^3$
- (C)  $y = \log(x^2 + x + 1)$
- (D)  $y = |x|$

(iii) Convert the implicit form into the explicit function :

(a)  $xy = 1$  (b)  $x^2y = 1$ .

## 10. APPLICATIONS OF FUNCTIONAL RULE :

**Illustration 13:** Determine all functions  $f$  satisfying the functional relation.

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)} \text{ where } x \in \mathbb{R} - \{0, 1\}$$

**Solution :** Given  $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$

$$= \frac{2}{x} - \frac{2}{1-x} \quad \dots(i)$$

Replacing  $x$  by  $\frac{1}{1-x}$  we obtain

$$f\left(\frac{1}{1-x}\right) + f\left(\frac{1}{1-\frac{1}{1-x}}\right) = 2(1-x) - \frac{2}{1-\frac{1}{1-x}}$$

$$\Rightarrow f\left(\frac{1}{1-x}\right) + f\left(1 - \frac{1}{x}\right) = -2x + \frac{2}{x} \quad \dots(ii)$$

Again replacing  $x$  by  $\left(1 - \frac{1}{x}\right)$  in (i) we obtain

$$\Rightarrow f\left(\frac{1}{1-x}\right) + f\left(\frac{1}{1-\left(1-\frac{1}{x}\right)}\right) = \frac{2}{1-\frac{1}{x}} - \frac{2}{1-\left(1-\frac{1}{x}\right)}$$

$$\Rightarrow f\left(1 - \frac{1}{x}\right) + f(x) = \frac{2x}{x-1} - 2x \quad \dots(iii)$$

subtracting (ii) from (i) then

$$f(x) - f\left(1 - \frac{1}{x}\right) = 2x - \frac{2}{1-x}$$

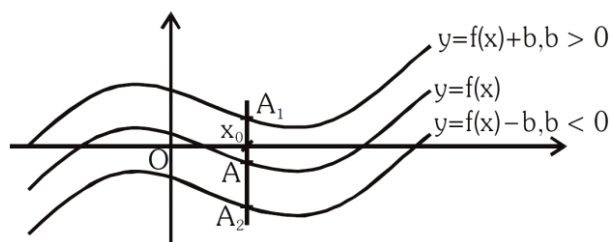
Now adding (iii) and (iv) we get

$$2f(x) = \frac{2x}{x-1} - \frac{2}{1-x}$$

$$\Rightarrow f(x) = \frac{x+1}{x-1}$$

## 11. BASIC TRANSFORMATIONS ON GRAPHS :

### (i) Drawing the graph of $y = f(x) + b$ , $b \in \mathbb{R}$ , from the known graph of $y = f(x)$



It is obvious that domain of  $f(x)$  and  $f(x) + b$  are the same. Let us take any point  $x_0$  in the domain of  $f(x)$ .  $y|_{x=x_0} = f(x_0)$ .

The corresponding point on  $f(x) + b$  would be  $f(x_0) + b$ .

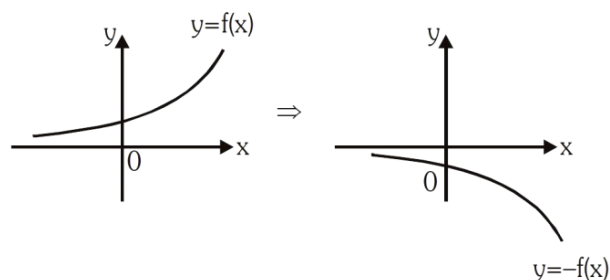
For  $b > 0 \Rightarrow f(x_0) + b > f(x_0)$  it means that the corresponding point on  $f(x) + b$  would be lying at a distance 'b' units above the point on  $f(x)$ .

For  $b < 0 \Rightarrow f(x_0) + b < f(x_0)$  it means that the corresponding point on  $f(x) + b$  would be lying at a distance 'b' units below the point on  $f(x)$ .

Accordingly the graph of  $f(x) + b$  can be obtained by translating the graph of  $f(x)$  either in the positive y-axis direction (if  $b > 0$ ) or in the negative y-axis direction (if  $b < 0$ ), through a distance  $|b|$  units.

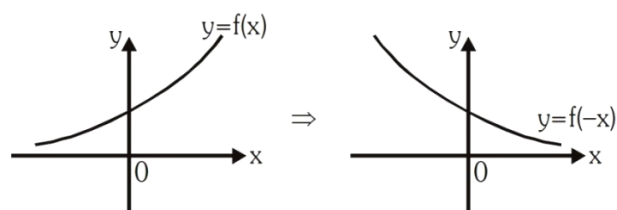
### (ii) Drawing the graph of $y = -f(x)$ from the known graph of $y = f(x)$

To draw  $y = -f(x)$ , take the image of the curve  $y = f(x)$  in the x-axis as plane mirror.



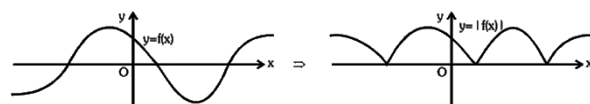
### (iii) Drawing the graph of $y = f(-x)$ from the known graph of $y = f(x)$

To draw  $y = f(-x)$ , take the image of the curve  $y = f(x)$  in the y-axis as plane mirror.



### (iv) Drawing the graph of $y = |f(x)|$ from the known graph of $y = f(x)$

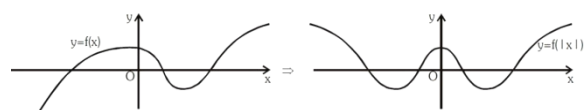
$|f(x)| = f(x)$  if  $f(x) \geq 0$  and  $|f(x)| = -f(x)$  if  $f(x) < 0$ . It means that the graph of  $f(x)$  and  $|f(x)|$  would coincide if  $f(x) \geq 0$  and for the portions where  $f(x) < 0$  graph of  $|f(x)|$  would be image of  $y = f(x)$  in x-axis.



### (v) Drawing the graph of $y = f(|x|)$ from the known graph of $y = f(x)$

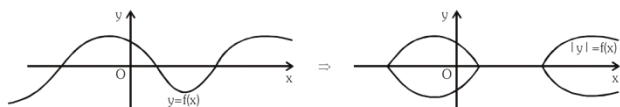
It is clear that,  $f(|x|) = \begin{cases} f(x), & x \geq 0 \\ f(-x), & x < 0 \end{cases}$ .

Thus  $f(|x|)$  would be an even function, graph of  $f(|x|)$  and  $f(x)$  would be identical in the first and the fourth quadrants (as  $x \geq 0$ ) and as such the graph of  $f(|x|)$  would be symmetric about the y-axis (as  $|x|$  is even).

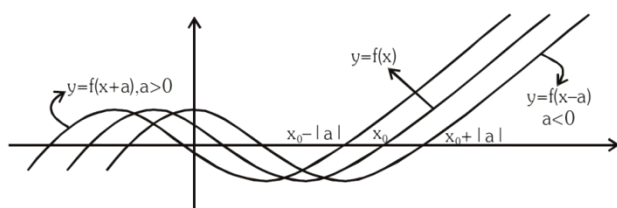


**(vi) Drawing the graph of  $|y| = f(x)$  from the known graph of  $y = f(x)$**

Clearly  $|y| \geq 0$ . If  $f(x) < 0$ , graph of  $|y| = f(x)$  would not exist. And if  $f(x) \geq 0$ ,  $|y| = f(x)$  would give  $y = \pm f(x)$ . Hence graph of  $|y| = f(x)$  would exist only in the regions where  $f(x)$  is non-negative and will be reflected about the x-axis only in those regions.

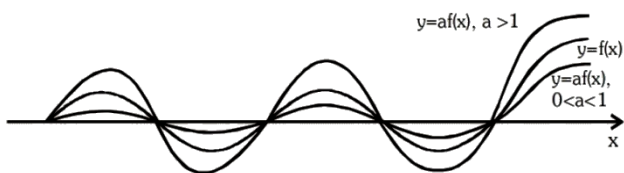


**(vii) Drawing the graph of  $y = f(x + a)$ ,  $a \in \mathbb{R}$  from the known graph of  $y = f(x)$**



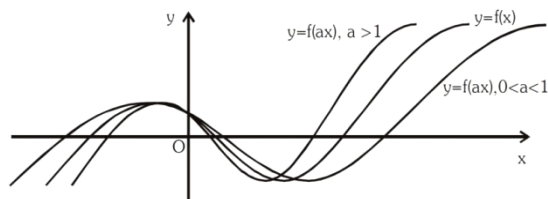
- If  $a > 0$ , shift the graph of  $f(x)$  through 'a' units towards left of  $f(x)$ .
- If  $a < 0$ , shift the graph of  $f(x)$  through 'a' units towards right of  $f(x)$ .

**(viii) Drawing the graph of  $y = af(x)$  from the known graph of  $y = f(x)$**



It is clear that the corresponding points (points with same x co-ordinates) would have their ordinates in the ratio of 1 : a.

**(ix) Drawing the graph of  $y = f(ax)$  from the known graph of  $y = f(x)$ .**



Let us take any point  $x_0 \in \text{domain of } f(x)$ . Let  $ax = x_0$  or  $x = \frac{x_0}{a}$ .

Clearly if  $0 < a < 1$ , then  $x > x_0$  and  $f(x)$  will stretch by  $\frac{1}{a}$  units along the x-axis and if  $a > 1$ ,  $x < x_0$ , then  $f(x)$  will compress by 'a' units along the x-axis.

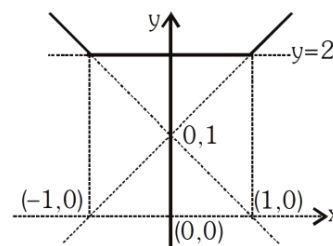
**Note :**

- A function  $h(x)$  is defined as  $h(x) = \max. \{f(x), g(x)\}$  then 
$$h(x) = \begin{cases} f(x) & f(x) \geq g(x) \\ g(x) & g(x) > f(x) \end{cases}$$
- A function  $h(x)$  is defined as  $h(x) = \min. \{f(x), g(x)\}$  then 
$$h(x) = \begin{cases} f(x) & f(x) \leq g(x) \\ g(x) & g(x) < f(x) \end{cases}$$

**Illustration 14 :** Find  $f(x) = \max \{1 + x, 1 - x, 2\}$ .

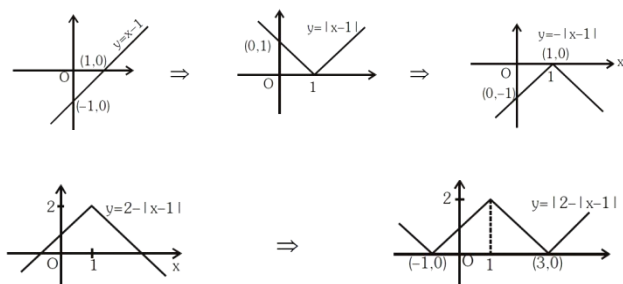
**Solution :** From the graph it is clear that

$$f(x) = \begin{cases} 1 - x & ; \quad x < -1 \\ 2 & ; \quad -1 \leq x \leq 1 \\ 1 + x & ; \quad x > 1 \end{cases}$$



**Illustration 15 :** Draw the graph of  $y = |2 - |x - 1||$ .

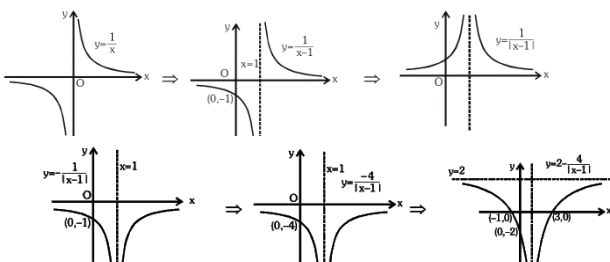
**Solution :**



**Illustration 16 :** Draw the graph of

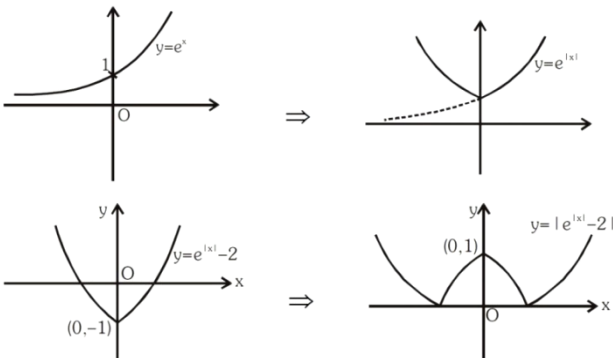
$$y = 2 - \frac{4}{|x-1|}$$

**Solution :**



**Illustration 17 :** Draw the graph of  $y = |e^{|x|} - 2|$

**Solution :**



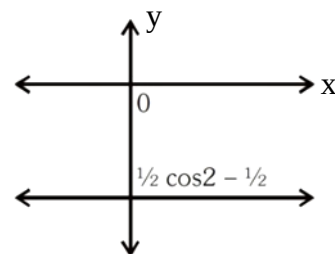
**Illustration 18 :** Draw the graph of

$$f(x) = \cos x \cos(x+2) - \cos^2(x+1).$$

**Solution :**  $f(x) = \cos x \cos(x+2) - \cos^2(x+1)$

$$= \frac{1}{2} [\cos(2x+2) + \cos 2] - \frac{1}{2} [\cos(2x+2) + 1]$$

$$= \frac{1}{2} \cos 2 - \frac{1}{2} < 0$$



### Do yourself -6

(i) Draw graph of following functions :

(a)  $y = |\ln|x|| + 1$

(b)  $y = \min.\{x^2 + 1, 3-x\}$

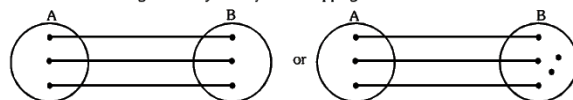
## 12. CLASSIFICATION OF FUNCTIONS :

### One-One Function (Injective mapping) :

A function  $f : A \rightarrow B$  is said to be a one-one function or injective mapping if different elements of  $A$  have different  $f$  images in  $B$ .

Thus there exist  $x_1, x_2 \in A$  &  $f(x_1), f(x_2) \in B$ ,  
 $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$  or  $x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)$ .

Diagrammatically an injective mapping can be shown as



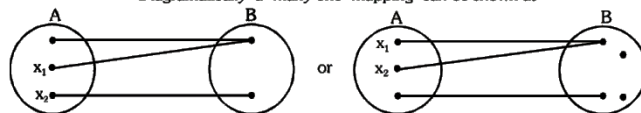
### Many-one function (not injective) :

A function  $f : A \rightarrow B$  is said to be a many one function if two or more elements of  $A$  have the same  $f$  image in  $B$ .

Thus  $f : A \rightarrow B$  is many one there exist

$x_1, x_2 \in A$ ,  $f(x_1) = f(x_2)$  but  $x_1 \neq x_2$ .

Diagrammatically a many one mapping can be shown as



**Note :**

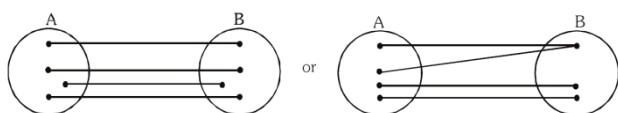
(i) If a line parallel to x-axis cuts the graph of the function at most at one point, then the function is one-one.

- (ii) If any line parallel to x-axis cuts the graph of the function atleast at two points, then  $f$  is many-one.
- (iii) If continuous function  $f(x)$  is always increasing or decreasing in whole domain, then  $f(x)$  is one-one.
- (iv) All linear functions are one-one.
- (v) All trigonometric functions in their domain are many one
- (vi) All even degree polynomials are many one
- (vii) Linear by Linear is one-one
- (viii) Quadratic by quadratic with no common factor is many one.

### Onto function (Surjective mapping) :

If the function  $f : A \rightarrow B$  is such that each element in  $B$  (co-domain) is the  $f$  image of atleast one element in  $A$ , then we say that  $f$  is a function from  $A$  'onto'  $B$ . Thus  $f : A \rightarrow B$  is surjective iff  $\forall b \in B, \exists$  some  $a \in A$  such that  $f(a) = b$ .

Diagrammatically surjective mapping can be shown as

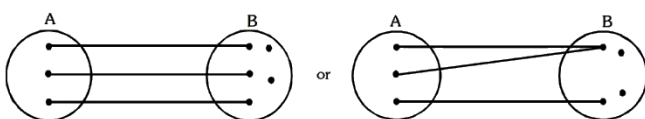


**Note that : if range is same as co-domain, then  $f(x)$  is onto.**

### Into function :

If  $f : A \rightarrow B$  is such that there exists atleast one element in co-domain which is not the image of any element in domain, then  $f(x)$  is into.

Diagrammatically into function can be shown as



### Note :

- (i) A polynomial function of degree even define from  $\mathbb{R} \rightarrow \mathbb{R}$  will always be into.
- (ii) A polynomial function of degree odd defined from  $\mathbb{R} \rightarrow \mathbb{R}$  will always be onto.
- (iii) Quadratic by quadratic without any common factor define from  $\mathbb{R} \rightarrow \mathbb{R}$  is always an into function.

**Thus a function can be one of these four types:**

- |       |                                                                                                |  |
|-------|------------------------------------------------------------------------------------------------|--|
| (i)   | one-one onto<br>(injective &<br>surjective)<br>(also known<br>as <b>Bijjective</b><br>mapping) |  |
| (ii)  | one-one into<br>(injective but<br>not surjective)                                              |  |
| (iii) | many-one<br>onto<br>(surjective but<br>not injective)                                          |  |
| (iv)  | many-one<br>into (neither<br>surjective nor<br>injective)                                      |  |

**Illustration 19 :** Let  $A = \{x : -1 \leq x \leq 1\} = B$  be a mapping  $f : A \rightarrow B$ . For each of the following functions from  $A$  to  $B$ , find whether it is bijective or non-bijective.

(a)  $f(x) = x|x|$       (b)  $f(x) = x^3$

(c)  $f(x) = \sin \frac{\pi x}{2}$

**Solution :**

(a)  $f(x) = x|x| = \begin{cases} -x^2 & , -1 < x < 0 \\ x^2 & , 0 \leq x < 1 \end{cases}$

Graphically,

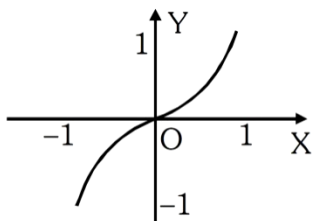
The graph shows  $f(x)$  is one-one, as the straight line parallel to x-axis cuts only at one point.

Here, range  $f(x) \in [-1, 1]$

Thus, range = co-domain

Hence, onto.

Therefore,  $f(x)$  is one-one onto or (Bijective).



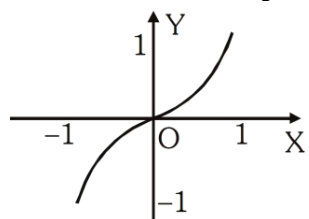
(b)  $f(x) = x^3$ ,

Graphically;

Graph shows  $f(x)$  is one-one onto (i.e.

Bijective)

[as explained in above example]

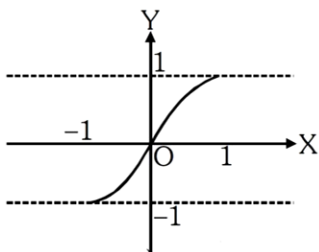


(c)  $f(x) = \sin \frac{\pi x}{2}$

Graphically;

Which shows  $f(x)$  is one-one and onto as range = co-domain.

Therefore,  $f(x)$  is bijective.



**Illustration 20 :** Let  $f : \mathbb{N} \rightarrow \mathbb{I}$  be a function defined as  $f(x) = x - 1000$ . Show that  $f(x)$  is an into function.

**Solution :** Let  $f(x) = y = x - 1000$

$\Rightarrow x = y + 1000 = g(y)$  (say)

here  $g(y)$  is defined for each  $y \in \mathbb{I}$ , but  $g(y) \notin \mathbb{N}$  for  $y \leq -1000$ .

Hence  $f(x)$  is into.

**Illustration 21 :** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function

defined by  $f(x) = x + \sqrt{x^2}$ , then  $f$  is

(A) injective (B) surjective

(C) bijective (D) None of these

**Solution :** We have,  $f(x) = x + \sqrt{x^2} = x + |x|$

Clearly,  $f$  is not one-one as  $f(-1) = f(-2) = 0$  and  $-1 \neq -2$

Also,  $f$  is not onto as  $f(x) \geq 0 \forall x \in \mathbb{R}$

$\therefore$  range of  $f = (0, \infty) \subset \mathbb{R}$

**Illustration 22 :** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function

defined as  $f(x) = 2x^3 + 6x^2 + 12x + 3 \cos x - 4 \sin x$ ; then  $f$  is -

(A) Injective (B) Surjective

(C) Bijective (D) Not Surjective

**Solution :** We have

$f(x) = 2x^3 + 6x^2 + 12x + 3 \cos x - 4 \sin x$

$\Rightarrow f'(x) = 6x^2 - 12x + 12 - 3 \sin x - 4 \cos x$

$f'(x) = \underbrace{6(x-1)^2 + 6}_{g(x)} - \underbrace{(3 \sin x + 4 \cos x)}_{h(x)}$

range of  $g(x) = [6, \infty)$

range of  $h(x) = [-5, 5]$

hence  $f'(x)$  always lies in the interval

$[1, \infty) \Rightarrow f'(x) > 0$

Hence  $f(x)$  is increasing i.e. one-one function

Now  $x \rightarrow \infty \Rightarrow f \rightarrow \infty$  &  $x \rightarrow -\infty \Rightarrow f \rightarrow -\infty$  &  $f(x)$  is continuous

hence its range is  $\mathbb{R} \Rightarrow f$  is onto so  $f$  is bijective.

**Ans. (C)**

**Illustration 23 :** Let  $f(x) = \frac{x^2 + 3x + a}{x^2 + x + 1}$ , where

$f : \mathbb{R} \rightarrow \mathbb{R}$ . Find the value of parameter 'a' so that the given function is one-one.

**Solution :**  $f(x) = \frac{x^2 + 3x + a}{x^2 + x + 1}$

$$f'(x) = \frac{(x^2 + x + 1)(2x + 3) - (x^2 + 3x + a)(2x + 1)}{(x^2 + x + 1)^2}$$

$$= \frac{-2x^2 + 2x(1 - a) + (3 - a)}{(x^2 + x + 1)^2}$$

Let,  $g(x) = -2x^2 + 2x(1 - a) + (3 - a)$

$g(x)$  will be negative if  $4(1 - a)^2 + 8(3 - a) < 0$

$\Rightarrow 1 + a^2 - 2a + 6 - 2a < 0 \Rightarrow (a - 2)^2 + 3 < 0$

which is not possible. Therefore function is not monotonic.

Hence, no value of a is possible.

**Do yourself -7**

(i) Is the function  $f : \mathbb{N} \rightarrow \mathbb{N}$  (the set of natural numbers) defined by  $f(x) = 2x + 3$  surjective?

(ii) Let  $A = \mathbb{R} - \{3\}$ ,  $B = \mathbb{R} - \{1\}$  and let  $f : A \rightarrow B$  defined by  $f(x) = \frac{x - 2}{x - 3}$ . Check whether the function  $f(x)$  is bijective or not.

(iii) A mapping  $f : A \rightarrow [-1, 1]$  defined by  $f(x) = \sin x$ ,  $\forall x \in \mathbb{R}$ , where A is a subset of  $\mathbb{R}$  (the set of all real numbers) is one-one and onto if A is the interval, then A belongs to

- (A)  $[0, 2\pi]$       (B)  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$   
(C)  $[-\pi, \pi]$       (D)  $[0, \pi]$

**13. COMPOSITE OF UNIFORMLY & NON-UNIFORMLY DEFINED FUNCTION:**

Let  $f : A \rightarrow B$  &  $g : B \rightarrow C$  be two functions. Then the function  $gof : A \rightarrow C$  defined by  $(gof)(x) = g(f(x)) \quad \forall x \in A$  is called the composite of the two functions  $f$  &  $g$ .

Diagrammatically  $\xrightarrow{x} \boxed{f} \xrightarrow{f(x)} \boxed{g} \longrightarrow g(f(x))$

Thus the image of every  $x \in A$  under the function  $gof$  is the g-image of f-image of x.

Note that  $gof$  is defined only if  $\forall x \in A$ ,  $f(x)$  is an element of the domain of 'g' so that we can take its g-image.

**Properties of composite functions:**

- (a) In general composite of functions is not commutative i.e.  $gof \neq fog$ .
- (b) The composition of functions is associative i.e. if  $f, g, h$  are three functions such that  $fo(goh)$  &  $(fog)oh$  are defined, then  $fo(goh) = (fog)oh$ .
- (c) The composition of two bijections is a bijection i.e. if  $f$  &  $g$  are two bijections such that  $gof$  is defined, then  $gof$  is also a bijection.

**Illustration 24 :**

If  $f(x) = x^2 + 1$ ,  $g(x) = \frac{1}{x - 1}$ , then find  $(fog)(x)$  and  $(gof)(x)$ .

**Solution :**

Given,  $f(x) = x^2 + 1$  ....(1)

$g(x) = \frac{1}{x - 1}$  .....(2)

Now  $(fog)(x) = f(g(x)) = f\left(\frac{1}{x - 1}\right) = f(z),$

$$\begin{aligned}\text{where } z &= \frac{1}{x-1} \\ &= z^2 + 1 \quad [\because f(x) = x^2 + 1] \\ &= \left(\frac{1}{x-1}\right)^2 + 1 = \frac{1}{(x-1)^2} + 1\end{aligned}$$

**Note :** Domain of  $f \circ g(x)$  is  $x \in \mathbb{R} - \{1\}$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x^2 + 1) = g(u), \\ \text{where } u &= x^2 + 1 \\ &= \frac{1}{u-1} = \frac{1}{x^2+1-1} = \frac{1}{x^2}\end{aligned}$$

**Note :** Domain if  $g \circ f(x)$  is  $x \in \mathbb{R} - \{0\}$

**Illustration 25 :**

If  $f$  be the greatest integer function and  $g$  be the modulus function, then

$$(g \circ f)\left(-\frac{5}{3}\right) - (f \circ g)\left(-\frac{5}{3}\right) =$$

- (A) 1 (B) -1  
(C) 2 (D) 4

**Solution :**

Given

$$(g \circ f)\left(-\frac{5}{3}\right) - (f \circ g)\left(-\frac{5}{3}\right) =$$

$$g\left\{f\left(-\frac{5}{3}\right)\right\} - f\left\{g\left(-\frac{5}{3}\right)\right\}$$

$$= g(-2) - f\left(\frac{5}{3}\right) = 2 - 1 = 1$$

**Ans.(A)**

**Illustration 26 :**

$$\text{Let } f(x) = \begin{cases} x+1, & x \leq 1 \\ 2x+1, & 1 < x \leq 2 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x^2, & -1 \leq x < 2 \\ x+2, & 2 \leq x \leq 3 \end{cases}, \text{ find } (f \circ g)$$

**Solution :**

$$f(g(x)) = \begin{cases} g(x)+1, & g(x) \leq 1 \\ 2g(x)+1, & 1 < g(x) \leq 2 \end{cases}$$

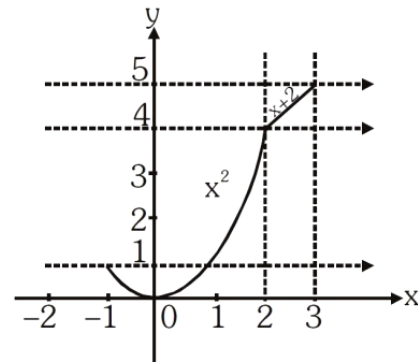
Here,  $g(x)$  becomes the variable that means we should draw the graph.

It is clear that  $g(x) \leq 1$  ;

$$\forall x \in [-1, 1] \text{ and } 1 < g(x) \leq 2 ;$$

$$\forall x \in (1, \sqrt{2}]$$

$$\Rightarrow f(g(x)) = \begin{cases} x^2+1, & -1 \leq x \leq 1 \\ 2x^2+1, & 1 < x \leq \sqrt{2} \end{cases}$$



**Illustration 27 :**

Find the domain and range of  $h(x) = g(f(x))$ , where

$$f(x) = \begin{cases} [x], & -2 \leq x \leq -1 \\ |x|+1, & -1 < x \leq 2 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} [x], & -\pi \leq x \leq 0 \\ \sin x, & 0 \leq x \leq \pi \end{cases}, \text{ } [.] \text{ denotes}$$

the greatest integer function.

**Solution :**

$$h(x) = g(f(x)) = \begin{cases} [f(x)], & -\pi \leq f(x) < 0 \\ \sin(f(x)), & 0 \leq f(x) \leq \pi \end{cases}$$

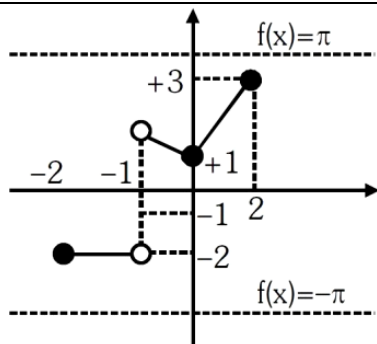
From graph of  $f(x)$ , we get

$$h(x) = \begin{cases} [[x]], & -2 \leq x \leq -1 \\ \sin(|x|+1), & -1 < x \leq 2 \end{cases}$$

$$\Rightarrow \text{Domain of } h(x) \text{ is } [-2, 2]$$

and Range of  $h(x)$  is

$$\{-2, -1\} \cup [\sin 3, 1]$$



**Do yourself -8 :**

- (i)  $f(x) = x^3 - x$  &  $g(x) = \sin 2x$ , find  
 (a)  $f(f(1))$  (b)  $f(f(-1))$   
 (c)  $f\left(g\left(\frac{\pi}{2}\right)\right)$  (d)  $f\left(g\left(\frac{\pi}{4}\right)\right)$   
 (e)  $g(f(1))$  (f)  $g\left(g\left(\frac{\pi}{2}\right)\right)$
- (ii) If  $f(x) = \begin{cases} x+1; & 0 \leq x < 2 \\ |x|; & 2 \leq x < 3 \end{cases}$ , then find  $f \circ g(x)$ .

**14. INVERSE OF A FUNCTION :**

Let  $f : A \rightarrow B$  be a one-one & onto function, then there exists a unique function  $g : B \rightarrow A$  such that  $f(x) = y \Leftrightarrow g(y) = x, \forall x \in A \text{ \& } y \in B$ . Then  $g$  is said to be inverse of  $f$ .

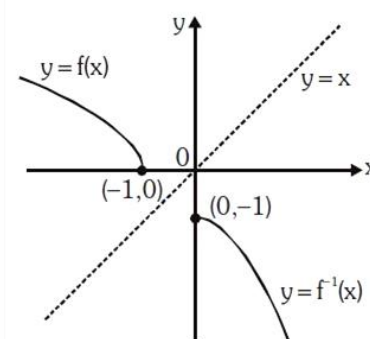
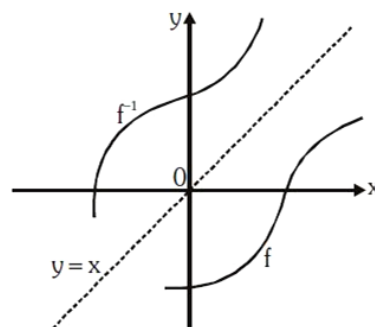
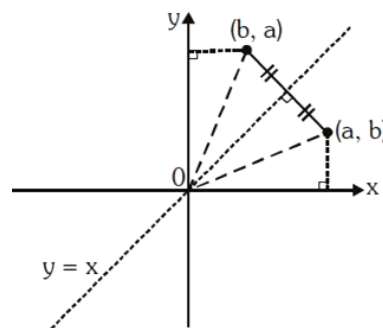
Thus  $g = f^{-1} : B \rightarrow A = \{(f(x), x) | (x, f(x)) \in f\}$ .

**Properties of inverse function :**

- (a) The inverse of a bijection is unique.  
 (b) If  $f : A \rightarrow B$  is a bijection &  $g : B \rightarrow A$  is the inverse of  $f$ , then  $f \circ g = I_B$  and  $g \circ f = I_A$ , where  $I_A$  &  $I_B$  are identity functions on the sets  $A$  &  $B$  respectively. If  $f \circ f = I$ , then  $f$  is inverse of itself.  
 (c) The inverse of a bijection is also a bijection.

(d) If  $f$  &  $g$  are two bijections  $f : A \rightarrow B$ ,  $g : B \rightarrow C$  then the inverse of  $g \circ f$  exists and  $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ .

(e) Since  $f(a) = b$  if and only if  $f^{-1}(b) = a$ , the point  $(a, b)$  is on the graph of ' $f$ ' if and only if the point  $(b, a)$  is on the graph of  $f^{-1}$ . But we get the point  $(b, a)$  from  $(a, b)$  by reflecting about the line  $y = x$ .



The graph of  $f^{-1}$  is obtained by reflecting the graph of  $f$  about the line  $y = x$ .

### Drawing the graph of $y = f^{-1}(x)$ from the known graph of $y = f(x)$

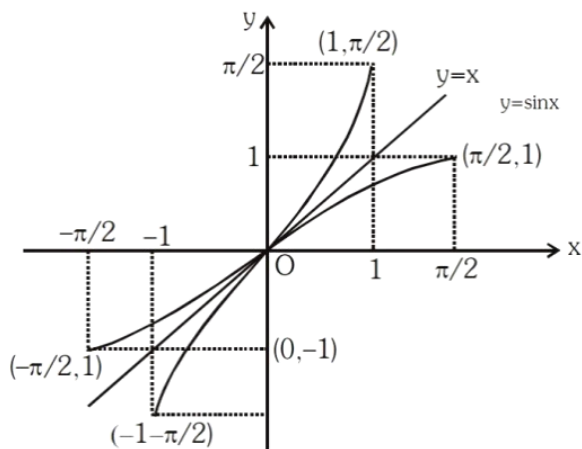
For drawing the graph of  $y = f^{-1}(x)$  take the reflection of  $y = f(x)$  about the line  $y = x$ . The reflected part would give us the graph of  $y = f^{-1}(x)$ .

e.g. let us draw the graph of  $y = \sin^{-1}x$ . We know that  $y = f(x) = \sin x$  is invertible if

$$f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

$\Rightarrow$  then inverse mapping would

$$f^{-1} : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$



### Illustration 28 :

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by  $f(x) = (e^x - e^{-x})/2$ . Is  $f(x)$  invertible? If so, find its inverse.

### Solution :

Let us check for invertibility of  $f(x)$  :

(a) One-One :

$$f(x) = \frac{1}{2}(e^x - e^{-x})$$

$$\Rightarrow f'(x) = \frac{1}{2}(e^x + e^{-x})$$

$\Rightarrow f'(x) > 0$ ,  $f(x)$  is increasing function

$\therefore f(x)$  is one-one function.

(b) Onto :

As  $x$  tends to larger and larger values so does  $f(x)$  and

when  $x \rightarrow \infty$ ,  $f(x) \rightarrow \infty$ .

Similarly as  $x \rightarrow -\infty$ ,  $f(x) \rightarrow -\infty$  i.e.  $-\infty < f(x) < \infty$  so long as  $x \in (-\infty, \infty)$

Hence the range of  $f$  is same as the set  $\mathbb{R}$ . Therefore  $f(x)$  is onto.

Since  $f(x)$  is both one-one and onto,  $f(x)$  is invertible.

(c) To find  $f^{-1}(x)$  : Interchange  $x$  &  $y$

$$\frac{e^y - e^{-y}}{2} = x \Rightarrow e^{2y} - 2xe^y - 1 = 0$$

$$\Rightarrow e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} \Rightarrow e^y = x \pm \sqrt{1+x^2}$$

Since  $e^y > 0$ , hence negative sign is ruled out and

$$\text{Hence } e^y = x + \sqrt{1+x^2}$$

Taking logarithm, we have  $y = \ln(x + \sqrt{1+x^2})$  or  $f^{-1}(x) = \ln(x + \sqrt{1+x^2})$

### Illustration 29 :

Find the inverse of the function  $f(x) = \log_a \left( x + \sqrt{x^2 + 1} \right)$ ;  $a > 1$  and assuming it to be an onto function.

### Solution :

$$\text{Given } f(x) = \log_a \left( x + \sqrt{x^2 + 1} \right)$$

$$\therefore f'(x) = \frac{\log_a e}{\sqrt{1+x^2}} > 0$$

which is a strictly increasing functions.

Thus,  $f(x)$  is injective, given that  $f(x)$  is onto. Hence the given function  $f(x)$  is invertible. Interchanging  $x$  &  $y$

$$\Rightarrow \log_a \left( y + \sqrt{(y)^2 + 1} \right) = x$$

$$\Rightarrow y + \sqrt{(y)^2 + 1} = a^x \quad \dots\dots(i)$$

$$\text{and } \sqrt{(y)^2 + 1} - y = a^{-x} \quad \dots\dots(ii)$$

From (i) and (ii), we get  $y = \frac{1}{2} (a^x - a^{-x})$

or  $f^{-1}(x) = \frac{1}{2} (a^x - a^{-x})$

**Illustration 30 :**

Find the inverse of the function  $f(x) = \ln(x^2 + 3x + 1)$ ;  $x \in [1, 3]$  and assuming it to be an onto function.

**Solution :**

Given  $f(x) = \ln(x^2 + 3x + 1)$

$$\therefore f'(x) = \frac{2x+3}{(x^2+3x+1)} > 0 \quad \forall x \in [1, 3]$$

Which is a strictly increasing function. Thus  $f(x)$  is injective, given that  $f(x)$  is onto. Hence the given function  $f(x)$  is invertible.

Interchanging  $x$  &  $y$

$$\Rightarrow (y)^2 + 3(y) + 1 - e^x = 0$$

$$\therefore y = \frac{-3 \pm \sqrt{9 - 4(1 - e^x)}}{2}$$

$$= \frac{-3 \pm \sqrt{(5 + 4e^x)}}{2}$$

$$\Rightarrow y = \frac{-3 + \sqrt{(5 + 4e^x)}}{2} \quad (\text{as } y \in [1, 3])$$

$$\text{Hence } f^{-1}(x) = \frac{-3 + \sqrt{(5 + 4e^x)}}{2}$$

**Do yourself -9 :**

- (i) Let  $f : [-1, 1] \rightarrow [-1, 1]$  defined by  $f(x) = x|x|$ , find  $f^{-1}(x)$ .
- (ii)  $f(x) = 1 + \ln(x + 2)$ , find  $f^{-1}(x)$ .

**15. ODD & EVEN FUNCTIONS :**

Consider a function  $f(x)$  such that both  $x$  and  $-x$  are in its domain then

If

$$\begin{cases} f(-x) = f(x) & \text{then } f \text{ is said to be an even function} \\ f(-x) = -f(x) & \text{then } f \text{ is said to be an odd function} \end{cases}$$

**Note :**

- (i)  $f(x) - f(-x) = 0 \Rightarrow f(x)$  is even &  $f(x) + f(-x) = 0 \Rightarrow f(x)$  is odd.
- (ii) A function may neither be odd nor even.
- (iii) The only function which is defined on the entire number line & is even and odd at the same time is  $f(x) = 0$ .
- (iv) Every constant function is even function.
- (v) Inverse of an even function is not defined.
- (vi) Every even function is symmetric about the  $y$ -axis & every odd function is symmetric about the origin.

**Special Note :** If a function  $f(x)$  is defined as  $f(a + x) = f(a - x)$  then this function is symmetric about line  $x = a$

- (vii) Every function which has ' $-x$ ' in it's domain whenever ' $x$ ' is in it's domain, can be expressed as the sum of an even & an odd function.

$$\text{i.e. } f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{EVEN}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{ODD}}$$

- (viii) If  $f(x)$  is odd and defined at  $x = 0$ . Then  $f(0) = 0$ .

| $f(x)$ | $g(x)$ | $f(x) + g(x)$        | $f(x) - g(x)$        | $f(x) \cdot g(x)$ | $f(x) / g(x)$ | $(gof)(x)$ | $(fog)(x)$ |
|--------|--------|----------------------|----------------------|-------------------|---------------|------------|------------|
| odd    | odd    | odd                  | odd                  | even              | even          | odd        | odd        |
| even   | even   | even                 | even                 | even              | even          | even       | even       |
| odd    | even   | neither odd nor even | neither odd nor even | odd               | odd           | even       | even       |
| even   | odd    | neither odd nor even | neither odd nor even | odd               | odd           | even       | even       |

**Illustration 31 :**

Which of the following functions is (are) even, odd or neither :

- (i)  $f(x) = x^2 \sin x$   
 (ii)  $f(x) = \sin x - \cos x$   
 (iii)  $f(x) = \frac{e^x + e^{-x}}{2}$

**Solution :**

- (i)  $f(-x) = (-x)^2 \sin(-x) = -x^2 \sin x = -f(x)$ .  
 Hence  $f(x)$  is odd.  
 (ii)  $f(-x) = \sin(-x) - \cos(-x) = -\sin x - \cos x$ .  
 Hence  $f(x)$  is neither even nor odd.  
 (iii)  $f(-x) = \frac{e^{-x} + e^{-(-x)}}{2} = \frac{e^{-x} + e^x}{2} = f(x)$   
 Hence  $f(x)$  even

**Illustration 32 :**

Identify the given functions as odd, even or neither :

- (i)  $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$   
 (ii)  $f(x + y) = f(x) + f(y)$  for all  $x, y \in \mathbb{R}$

**Solution :**

- (i)  $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$   
 Clearly domain of  $f(x)$  is  $\mathbb{R} \sim \{0\}$ . We have,  

$$f(-x) = \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1 = \frac{-e^x \cdot x}{1 - e^x} - \frac{x}{2} + 1$$

$$= \frac{(e^x - 1 + 1)x}{(e^x - 1)} - \frac{x}{2} + 1$$

$$= x + \frac{x}{e^x - 1} - \frac{x}{2} + 1 = \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x)$$

Hence  $f(x)$  is an even function.

- (ii)  $f(x + y) = f(x) + f(y)$  for all  $x, y \in \mathbb{R}$   
 Replacing  $x, y$  by zero, we get  
 $f(0) = 2f(0)$   
 $\Rightarrow f(0) = 0$

Replacing  $y$  by  $-x$ , we get

$$f(x) + f(-x) = f(0) = 0$$

$$\Rightarrow f(x) = -f(-x)$$

Hence  $f(x)$  is an odd function.

**Do yourself -10 :**

- (i) Which of the following functions is (are) even, odd or neither :  
 (a)  $f(x) = x^3 \sin 3x$   
 (b)  $f(x) = \frac{e^{x^2} + e^{-x^2}}{2x}$   
 (c)  $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$   
 (d)  $f(x) = x^2 + 2^x$

**16. PERIODIC FUNCTION :**

A function  $f(x)$  is called periodic if there exists a positive number  $T$  ( $T > 0$ ), where  $T$  is the smallest such value called the period of the function such that  $f(x + T) = f(x)$ , for all values of  $x$ ,  $x + T$  within the domain of  $f$ .

**Note :**

- (i) Odd powers of  $\sin x$ ,  $\cos x$ ,  $\sec x$ ,  $\csc x$  are periodic with period  $2\pi$ .

- (ii) Non zero integral powers of  $\tan x$ ,  $\cot x$  are periodic with period  $\pi$ .
- (iii) Non zero even powers or modulus of  $\sin x$ ,  $\cos x$ ,  $\sec x$ ,  $\csc x$  are periodic with period  $\pi$ .
- (iv)  $f(T) = f(0) = f(-T)$ , where 'T' is the period.
- (v) If  $f(x)$  has a period T then  **$f(ax + b)$  has a period  $T/|a|$  ( $a \neq 0$ ).**

**Proof :** Let  $f(x + T) = f(x)$  and  $f[a(x + T)] = f(ax + b)$

$$f(ax + b + aT) = f(ax + b)$$

$$f(y + aT) = f(y) = f(y + T)$$

$$\Rightarrow T = aT' \Rightarrow T' = \frac{T}{a}$$

- (vi) If  $f(x)$  &  $g(x)$  are periodic with period  $T_1$  &  $T_2$  respectively, then period of  $f(x) \pm g(x)$  is L.C.M. of  $(T_1, T_2)$ .
- (a) LCM of  $T_1$  &  $T_2$  is defined when  $T_1/T_2$  is rational.
- (b)  $\text{LCM of } \left\{ \frac{a}{b}, \frac{p}{q} \right\} = \frac{\text{LCM of } (a, p)}{\text{HCF of } (b, q)}$
- In case if there exists a finite K such that  $K < \text{LCM of } T_1 \text{ and } T_2$  and overall function repeats itself after every K, then period of the function will be K.
- (vii) Every constant function is always periodic, whose period is undefined.
- (viii) Inverse of a periodic function does not exist.

### Illustration 33 :

Find the periods (if periodic) of the following function, where  $[.]$  denote the greatest integer function

- (i)  $f(x) = e^{\ln(\sin x)} + \tan^3 x - \csc(3x - 5)$
- (ii)  $f(x) = x - [x - b]$ ,  $b \in \mathbb{R}$

- (iii)  $f(x) = \frac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$
- (iv)  $f(x) = \tan \frac{\pi}{2}[x]$
- (v)  $f(x) = \cos(\sin x) + \cos(\cos x)$
- (vi)  $f(x) = \frac{(1 + \sin x)(1 + \sec x)}{(1 + \cos x)(1 + \csc x)}$
- (vii)  $f(x) = e^{x - [x] + |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi|}$

### Solution :

(i)  $f(x) = e^{\ln(\sin x)} + \tan^3 x - \csc(3x - 5)$

Period of  $e^{\ln(\sin x)} = 2\pi$ ,  $\tan^3 x = \pi$

$$\csc(3x - 5) = \frac{2\pi}{3}$$

$$\therefore \text{Period} = 2\pi$$

$$(vi) \quad f(x) = \frac{(1 + \sin x)(1 + \sec x)}{(1 + \cos x)(1 + \operatorname{cosec} x)}$$

$$= \frac{(1 + \sin x)(1 + \cos x) \sin x}{\cos x(1 + \sin x)(1 + \cos x)}$$

$$\Rightarrow f(x) = \tan x$$

Hence  $f(x)$  has period  $\pi$ .

$$(vii) \quad f(x) = e^{x - [x] + |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi|}$$

Period of  $x - [x] = 1$

Period of  $|\cos \pi x| = 1$

Period of  $|\cos 2\pi x| = \frac{1}{2}$

.....

Period of  $|\cos n\pi x| = \frac{1}{n}$

So period of  $f(x)$  will be L.C.M. of all period = 1

#### Illustration 34 :

Find the periods (if periodic) of the following functions, where  $[.]$  denotes the greatest integer function

$$(i) \quad f(x) = e^{x - [x]} + \sin x$$

$$(ii) \quad f(x) = \sin \frac{\pi x}{\sqrt{2}} + \cos \frac{\pi x}{\sqrt{3}}$$

$$(iii) \quad f(x) = \sin \frac{\pi x}{\sqrt{3}} + \cos \frac{\pi x}{2\sqrt{3}}$$

#### Solution :

$$(i) \quad \text{Period of } e^{x - [x]} = 1$$

Period of  $\sin x = 2\pi$

$\therefore$  L.C.M. of rational and an irrational number does not exist.

$\therefore$  not periodic.

$$(ii) \quad \text{Period of } \sin \frac{\pi x}{\sqrt{2}} = \frac{2\pi}{\pi/\sqrt{2}} = 2\sqrt{2}$$

$$\text{Period of } \cos \frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$$

$\therefore$  L.C.M. of two different kinds of irrational number does not exist.

$\therefore$  not periodic.

$$(iii) \quad \text{Period of } \sin \frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$$

$$\text{Period of } \cos \frac{\pi x}{2\sqrt{3}} = \frac{2\pi}{\pi/2\sqrt{3}} = 4\sqrt{3}$$

$\therefore$  L.C.M. of two similar irrational number exist.

$\therefore$  Periodic with period =  $4\sqrt{3}$

**Ans.**

#### Do yourself -11 :

(i) Find the periods (if periodic) of the following functions.

$$(a) \quad f(x) = \ln(\cos x) + \tan^3 x.$$

$$(b) \quad f(x) = e^{x - [x]}, [.] \text{ denotes greatest integer function}$$

$$(c) \quad f(x) = \left| \sin \frac{x}{2} \right| + \left| \cos \frac{x}{2} \right|$$

#### 17. GENERAL :

If  $x, y$  are independent variables, then :

$$(a) \quad f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$$

$$(b) \quad f(xy) = f(x) \cdot f(y) \Rightarrow f(x) = x^n, n \in \mathbb{R} \text{ or } f(x) = 0$$

$$(c) \quad f(x + y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx} \text{ or } f(x) = 0$$

$$(d) \quad f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx, \text{ where } k \text{ is a constant.}$$

**Miscellaneous Illustration :**

**Illustration 35 :**

ABCD is a square of side  $\ell$ . A line parallel to the diagonal BD at a distance 'x' from the vertex A cuts two adjacent sides. Express the area of the segment of the square with A at a vertex, as a function of x. Find this area at  $x = 1/\sqrt{2}$  and at  $x = 2$ , when  $\ell = 2$ .

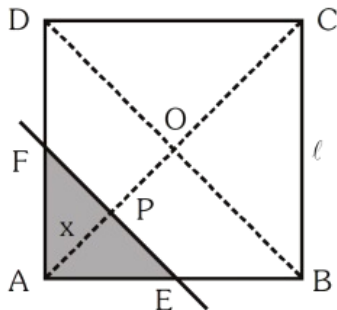
**Solution :**

There are two different situations

**Case-I :** when  $x = AP \leq OA$ , i.e.,  $x \leq \frac{\ell}{\sqrt{2}}$

$$\text{ar}(\triangle AEF) = \frac{1}{2} x \cdot 2x = x^2$$

$$(\because PE = PF = PA = x)$$



**Case-II :** when  $x = AP > OA$ , i.e.,  $x > \frac{\ell}{\sqrt{2}}$  but

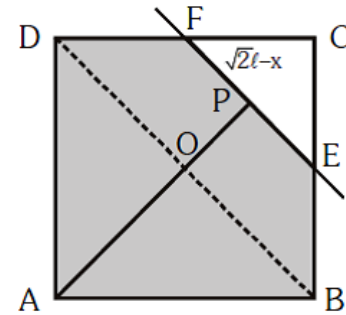
$$x \leq \sqrt{2}\ell$$

$$\begin{aligned} \text{ar}(\text{ABEFDA}) &= \text{ar}(\text{ABCD}) - \text{ar}(\triangle CFE) \\ &= \ell^2 - \frac{1}{2}(\sqrt{2}\ell - x) \cdot 2(\sqrt{2}\ell - x) \end{aligned}$$

$$[\because CP = \sqrt{2}\ell - x]$$

$$= \ell^2 - (\ell^2 + x^2 - 2\sqrt{2}\ell x) = 2\sqrt{2}\ell x - x^2 - \ell^2$$

$\therefore$  the required function  $s(x)$  is as follows :



$$s(x) = \begin{cases} x^2, & 0 \leq x \leq \frac{\ell}{\sqrt{2}} \\ 2\sqrt{2}\ell x - x^2 - \ell^2, & \frac{\ell}{\sqrt{2}} < x \leq \sqrt{2}\ell \end{cases};$$

$$\text{area of } s(x) = \begin{cases} \frac{1}{2} & \text{at } x = \frac{1}{\sqrt{2}} \\ 8(\sqrt{2} - 1) & \text{at } x = 2 \end{cases}$$

**Ans.**

**Illustration 36 :**

If the function  $f(x)$  satisfies the functional rule,  $f(x + y) = f(x) + f(y) \forall x, y \in \mathbb{R}$  &  $f(1) = 5$ , then find  $\sum_{n=1}^m f(n)$

**Solution :**

Here,  $f(x + y) = f(x) + f(y)$ ;

put  $x = t - 1, y = 1$

$$f(t) = f(t - 1) + f(1) \dots (1)$$

$$\therefore f(t) = f(t - 1) + 5$$

$$\Rightarrow f(t) = \{f(t - 2) + 5\} + 5$$

$$\Rightarrow f(t) = f(t - 2) + 2(5)$$

$$\Rightarrow f(t) = f(t - 3) + 3(5)$$

.....

.....

$$\Rightarrow f(t) = f\{t - (t - 1)\} + (t - 1)5$$

$$\Rightarrow f(t) = f(1) + (t - 1)5$$

$$\Rightarrow f(t) = 5 + (t - 1)5$$

$$\Rightarrow f(t) = 5t$$

$$\therefore \sum_{n=1}^m f(n) = \sum_{n=1}^m (5n) = 5[1 + 2 + 3 + \dots +$$

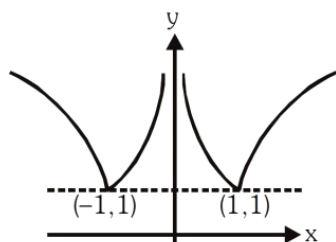
$$m] = \frac{5m(m+1)}{2}$$

$$\text{Hence, } \sum_{n=1}^m f(n) = \frac{5m(m+1)}{2}.$$

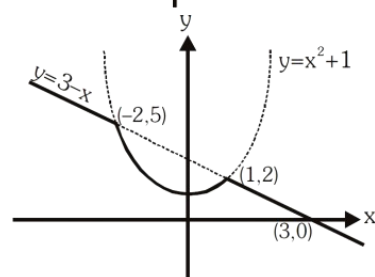
### ANSWERS FOR DO YOURSELF

- 1 : (i) (a)  $x \in (0, \infty)$   
 (b)  $x \in (-\infty, 0) \cup (4, \infty)$   
 (ii) (a)  $\left[\frac{1}{2}, \infty\right)$  (b)  $[-1, 1]$   
 (c)  $[-2, 2]$  (d)  $[-1, 1]$
- 2 : (i) (A)  $\rightarrow (p, q, r, s)$ , (B)  $\rightarrow (p)$ , (C)  $\rightarrow (t)$ , (D)  $\rightarrow (r)$
- 3 : (i) (a)  $(-1, 0) \cup (0, 1]$   
 (b)  $\left(-\infty, -\frac{1}{2}\right] \cup (0, 1) \cup (2, \infty)$   
 (ii) (a)  $(-\infty, \infty)$  (b)  $\left[\frac{1}{3}, 1\right]$
- 4 : (i) (a) no (b) no (c) yes
- 5 : (i)  $\left[0, \frac{1}{2}\right]$  (ii) A  
 (iii) (a)  $y = \frac{1}{x}$  (b)  $y = \frac{1}{x^2}$

6 : (i)



(ii)



7 : (i) not onto (ii) yes (iii) B

8 : (i) (a) 0 (b) 0 (c) 0  
 (d) 0 (e) 0 (f) 0

$$(ii) \begin{cases} x+2, & 0 \leq x < 1 \\ x+1, & 1 \leq x < 2 \\ x, & 2 \leq x < 3 \end{cases}$$

$$9 : (i) f^{-1}(x) = \begin{cases} -\sqrt{-x}, & -1 \leq x \leq 0 \\ \sqrt{x}, & 0 \leq x \leq 1 \end{cases}$$

$$(ii) y = -2 + e^{x-1}$$

10 : (i) (a) even (b) odd (c) odd  
 (d) neither even nor odd

11. (i) (a)  $2\pi$  (b) 1 (c)  $\pi$

## EXERCISE # 1

### Subjective Type Questions

1. Find the domain of definition of the given functions :

(i)  $y = \sqrt{-px}$  ( $p > 0$ )

(ii)  $y = \frac{1}{x^2 + 1}$

(iii)  $y = \frac{1}{x^3 - x}$

(iv)  $y = \frac{1}{\sqrt{x^2 - 4x}}$

(v)  $y = \sqrt{x^2 - 4x + 3}$

(vi)  $y = \frac{x}{\sqrt{x^2 - 3x + 2}}$

(vii)  $y = \sqrt{1 - |x|}$

(vii)  $y = \log_x 2$

(ix)  $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$

(x)  $y = \sqrt{x} + \sqrt[3]{\frac{1}{x-2}} - \log_{10}(2x-3)$

(xi)  $y = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$

(xii)  $y = \frac{1}{\sqrt{\sin x}} + \sqrt[3]{\sin x}$

(xiii)  $y = \log_{10}(\sqrt{x-4} + \sqrt{6-x})$

(xiv)  $y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$

2. Find the range of the following functions :

(i)  $f(x) = \frac{x-1}{x+2}$

(ii)  $f(x) = \frac{2}{x}$

(iii)  $f(x) = \frac{1}{x^2 - x + 1}$

(iv)  $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$

(v)  $f(x) = e^{(x-1)^2}$

(vi)  $f(x) = x^3 - x^2 + x + 1$

(vii)  $f(x) = \log(x^8 + x^4 + x^2 + 1)$

(viii)  $f(x) = \sin^2 x - 2\sin x + 4$

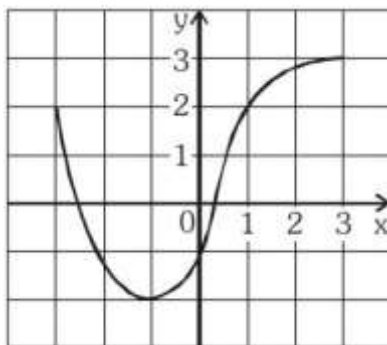
(ix)  $f(x) = \sin(\log_2 x)$

(x)  $f(x) = 2^{x^2} + 1$

(xi)  $f(x) = \frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1}$

(xii)  $f(x) = \frac{1}{8 - 3\sin x}$

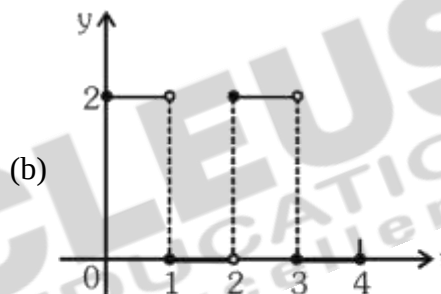
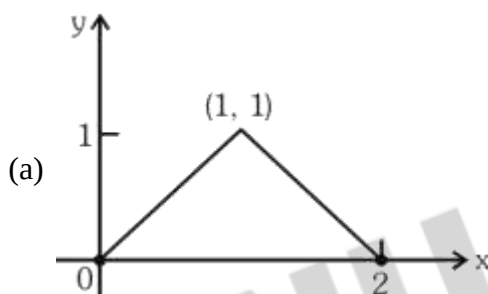
3. The graph of a function  $f$  is given.



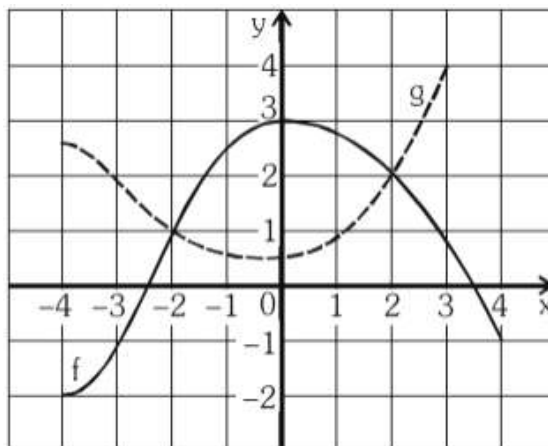
- (a) State the value of  $f(-1)$ .  
 (b) For what values of  $x$  is  $f(x) = 2$ ?  
 (c) State the domain and range of  $f$ .  
 (d) On what interval is  $f$  increasing?  
 (e) Estimated value of  $f(2)$  is :  
 (A) 2.2                      (B) 2.8                      (C) 2.5                      (D) 3  
 (f) Estimated value of  $x$  such that  $f(x) = 0$ , is :  
 (A) -2.5                      (B) 0.8                      (C) -2.9                      (D) 0.3

4. Graph the function  $F(x) = \begin{cases} 3-x, & x \leq 1 \\ 2x, & x > 1 \end{cases}$

5. Find a formula for each function graphed



6. The graphs of  $f$  and  $g$  are given.  
 (a) State the value of  $f(-4)$  and  $g(3)$ .  
 (b) For what value of  $x$  is  $f(x) = g(x)$ ?  
 (c) Estimate the solution of the equation  $f(x) = -1$ .  
 (d) On what interval is  $f$  decreasing?  
 (e) State the domain and range of  $f$ .  
 (f) State the domain and range of  $g$ .



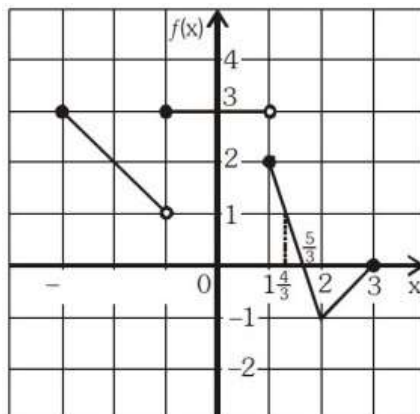
7. Solve the following inequalities using graph of  $f(x)$  :

(a)  $0 \leq f(x) \leq 1$

(b)  $-1 \leq f(x) \leq 2$

(c)  $2 \leq f(x) \leq 3$

(d)  $f(x) > -1$  &  $f(x) < 0$



### Straight Objective Type

8. If  $[a]$  denotes the greatest integer less than or equal to  $a$  and  $-1 \leq x < 0$ ,  $0 \leq y < 1$ ,  $1 \leq z < 2$ ,

then  $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$  is equal to –

(A)  $[x]$

(B)  $[y]$

(C)  $[z]$

(D) none of these

9. If  $[x]$  and  $\{x\}$  denotes the greatest integer function less than or equal to  $x$  and fractional part function respectively, then the number of real  $x$ , satisfying the equation  $(x-2)[x] = \{x\} - 1$ , is-

(A) 0

(B) 1

(C) 2

(D) infinite

10. The range of the function  $f(x) = \operatorname{sgn}\left(\frac{\sin^2 x + 2\sin x + 4}{\sin^2 x + 2\sin x + 3}\right)$  is (where  $\operatorname{sgn}(\cdot)$  denotes signum function)-

(A)  $\{-1, 0, 1\}$

(B)  $\{-1, 0\}$

(C)  $\{1\}$

(D)  $\{0, 1\}$

11. If  $2f(x) - 3f\left(\frac{1}{x}\right) = x^2$ ,  $x$  is not equal to zero, then  $f(2)$  is equal to-

(A)  $-\frac{7}{4}$

(B)  $\frac{5}{2}$

(C)  $-1$

(D) none of these

12. The number of integers lying in the domain of the function  $f(x) = \sqrt{\log_{0.5}\left(\frac{5-2x}{x}\right)}$  is -

(A) 3

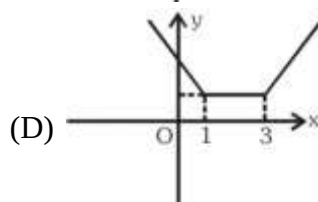
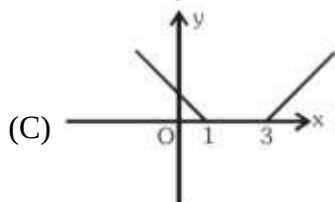
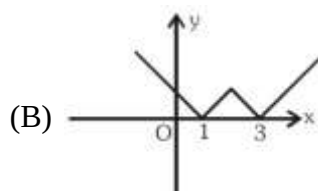
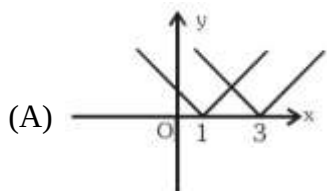
(B) 2

(C) 1

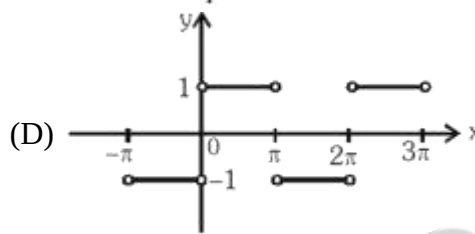
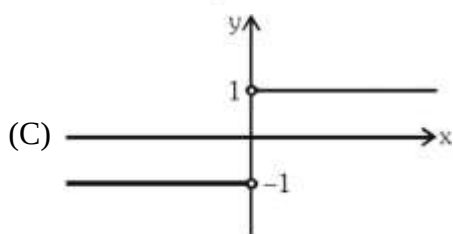
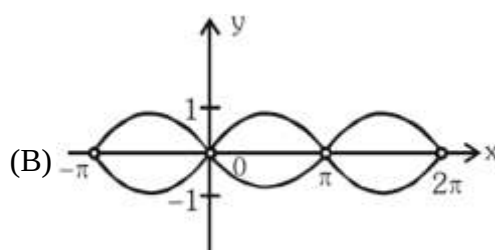
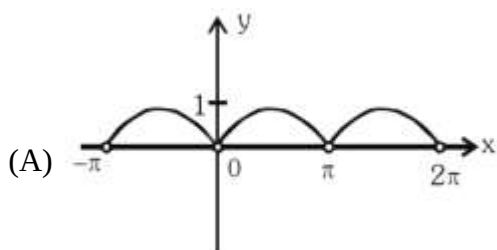
(D) 0

13. The range of the function  $f : \mathbb{N} \rightarrow \mathbb{I}; f(x) = (-1)^{x-1}$ , is -  
 (A)  $[-1, 1]$  (B)  $\{-1, 1\}$  (C)  $\{0, 1\}$  (D)  $\{0, 1, -1\}$
14. The range of the function  $f(x) = e^{-x} + e^x$ , is -  
 (A)  $f(x) \geq 1$  (B)  $f(x) \leq 1$  (C)  $f(x) \geq 2$  (D)  $f(x) \leq 2$
15. If  $f(x) = \frac{4^x}{4^x + 2}$ , then  $f(x) + f(1-x)$  is equal to-  
 (A) 0 (B) -1 (C) 1 (D) 4
16. The range of the function  $f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$  is  
 (A)  $[\sqrt{3}, \sqrt{7}]$  (B)  $[\sqrt{3}, \sqrt{5}]$  (C)  $[\sqrt{2}, \sqrt{3}]$  (D)  $[\sqrt{3}, \sqrt{6}]$
17. A function  $f$  has domain  $[-1, 2]$  and range  $[0, 1]$ . The domain and range respectively of the function  $g$  defined by  $g(x) = 1 - f(x+1)$  is  
 (A)  $[-1, 1]; [-1, 0]$  (B)  $[-2, 1]; [0, 1]$  (C)  $[0, 2]; [-1, 0]$  (D)  $[1, 3]; [-1, 0]$
18. For the function  $f(x) = \frac{e^x + 1}{e^x - 1}$ , if  $n(d)$  denotes the number of integers which are not in its domain and  $n(r)$  denotes the number of integers which are not in its range, then  $n(d) + n(r)$  is equal to -  
 (A) 2 (B) 3 (C) 4 (D) Infinite
19. If  $x^4 f(x) - \sqrt{1 - \sin 2\pi x} = |f(x)| - 2f(x)$ , then  $f(-2)$  equals  
 (A)  $\frac{1}{17}$  (B)  $\frac{1}{11}$  (C)  $\frac{1}{19}$  (D) 0
20. Let  $f : \mathbb{R} - \left\{ \frac{-15}{2} \right\} \rightarrow \mathbb{R} - \left\{ \frac{1}{2} \right\}$  be defined by  $f(x) = \frac{x+10}{2x+15}$  then  $f(x)$  is -  
 (A) one-one but not onto (B) many one but not-onto  
 (C) one-one and onto (D) many one and onto
21.  $f : \mathbb{R} \rightarrow \mathbb{R} f(x) = \frac{2x^2 - 5x + 3}{8x^2 + 9x + 11}$ , then  $f$  is -  
 (A) one-one onto (B) many-one onto  
 (C) one-one into (D) many one into

22. If  $f : \mathbb{R} \rightarrow \mathbb{R}$  &  $f(x) = \frac{\sin([x]\pi)}{x^2 + 2x + 3} + 2x - 1 + \sqrt{x(x-1) + \frac{1}{4}}$  (where  $[x]$  denotes integral part of  $x$ ), then  $f(x)$  is -  
 (A) one-one but not onto (B) one-one & onto  
 (C) onto but not one-one (D) neither one-one nor onto
23. Which of the following function is surjective but not injective  
 (A)  $f : \mathbb{R} \rightarrow \mathbb{R} f(x) = x^4 + 2x^3 - x^2 + 1$  (B)  $f : \mathbb{R} \rightarrow \mathbb{R} f(x) = x^3 + x + 1$   
 (C)  $f : \mathbb{R} \rightarrow \mathbb{R}^+ f(x) = \sqrt{1+x^2}$  (D)  $f : \mathbb{R} \rightarrow \mathbb{R} f(x) = x^3 + 2x^2 - x + 1$
24. If  $f(x) = x|x|$  then  $f^{-1}(x)$  equals-  
 (A)  $\sqrt{|x|}$  (B)  $(\text{sgn } x) \cdot \sqrt{|x|}$  (C)  $-\sqrt{|x|}$  (D) Does not exist  
 (where  $\text{sgn}(x)$  denotes signum function of  $x$ )
25. If  $f : (-\infty, 3] \rightarrow [7, \infty)$ ;  $f(x) = x^2 - 6x + 16$ , then which of the following is true -  
 (A)  $f^{-1}(x) = 3 + \sqrt{x-7}$  (B)  $f^{-1}(x) = 3 - \sqrt{x-7}$   
 (C)  $f^{-1}(x) = \frac{1}{x^2 - 6x + 16}$  (D)  $f$  is many-one
26.  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that  $f(x) = \ln(x + \sqrt{x^2 + 1})$ . Another function  $g(x)$  is defined such that  $g \circ f(x) = x, \forall x \in \mathbb{R}$ . Then  $g(2)$  is -  
 (A)  $\frac{e^2 + e^{-2}}{2}$  (B)  $e^2$  (C)  $\frac{e^2 - e^{-2}}{2}$  (D)  $e^{-2}$
27. Let  $P(x) = kx^3 + 2k^2x^2 + k^3$ . The sum of all real numbers  $k$  for which  $(x-2)$  is a factor of  $P(x)$ , is  
 (A) 4 (B) 8 (C) -4 (D) -8
28. Which of the following is the graph of  $y = |x-1| + |x-3|$ ?



29. Which of the following is the graph of  $y = \frac{|\sin x|}{\sin x}$  ?



30. Period of function  $f(x) = \min\{\sin x, |x|\} + \frac{x}{\pi} - \left[\frac{x}{\pi}\right]$  (where  $[.]$  denotes greatest integer function) is -  
(A)  $\pi/2$  (B)  $\pi$  (C)  $2\pi$  (D)  $4\pi$

31.  $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$  then let us define a function  $f(x) = \det. (A^T A^{-1})$  then which of the following can not be the value of  $\underbrace{f(f(f(\dots f(x))))}_{n \text{ times}}$  is ( $n \geq 2$ )  
(A)  $f^n(x)$  (B) 1 (C)  $f^{n-1}(x)$  (D)  $nf(x)$

32. The number of integral values of  $x$  satisfying the inequality  $[x - 5][x - 3] + 2 < [x - 5] + 2[x - 3]$  (where  $[.]$  represents greatest integer function) is -  
(A) 0 (B) 1 (C) 2 (D) 3

33. Range of function  $f(x) = \log_2 \left( \frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$  is given by

- (A)  $(0, \infty)$  (B)  $\left[\frac{1}{2}, 1\right]$  (C)  $[1, 2]$  (D)  $\left[\frac{1}{4}, 1\right]$

34. A lion moves in the region given by the graph  $y - |y| - x + |x| = 0$ . Then on which of the following curve a person can move so that he does not encounter lion -

- (A)  $y = e^{-|x|}$  (B)  $y = \frac{1}{x}$  (C)  $y = \text{signum}(x)$  (D)  $y = -|4 + |x||$

35. Suppose,  $f(x, n) = \sum_{k=1}^n \log_x \left( \frac{k}{x} \right)$ , then the value of  $x$  satisfying the equation  $f(x, 10) = f(x, 11)$ , is  
(A) 9 (B) 10 (C) 11 (D) none
36. Range of  $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}$ ;  $x \in \left( 0, \frac{\pi}{2} \right)$  is -  
(A) (0,1) (B) (1,∞) (C) (-1, 0) (D) (-∞,-1)
37. If  $f(x, y) = \max(x, y) + \min(x, y)$  and  $g(x, y) = \max(x, y) - \min(x, y)$ , then the value of  $f\left(g\left(-\frac{2}{3}, -\frac{3}{2}\right), g(-3, -4)\right)$  is greater than -  
(A) 1 (B) 2 (C) 3 (D) 4
38. If functions  $f(x)$  and  $g(x)$  are defined on  $\mathbb{R} \rightarrow \mathbb{R}$  such that  $f(x) = \begin{cases} x+3, & x \in \text{rational} \\ 4x, & x \in \text{irrational} \end{cases}$ ,  
 $g(x) = \begin{cases} x+\sqrt{5}, & x \in \text{irrational} \\ -x, & x \in \text{rational} \end{cases}$  then  $(f-g)(x)$  is -  
(A) one-one & onto (B) neither one-one nor onto  
(C) one-one but not onto (D) onto but not one-one
39. Let  $f : A \rightarrow B$  be an onto function such that  $f(x) = \sqrt{x-2-2\sqrt{x-3}} - \sqrt{x-2+2\sqrt{x-3}}$  then set 'B' is -  
(A) [-2,0] (B) [0,2] (C) [-3,0] (D) [-1,0]
40. Let  $f(x) = \frac{x}{1-x}$  and let  $\alpha$  be a real number. If  $x_0 = \alpha$ ,  $x_1 = f(x_0)$ ,  $x_2 = f(x_1)$ , ..... &  $x_{2011} = -\frac{1}{2012}$  then the value of  $\alpha$  is -  
(A)  $\frac{2011}{2012}$  (B) 1 (C) 2011 (D) -1
41. If  $f_1(x) = 2^{f_2(x)}$ , where  $f_2(x) = 2012^{f_3(x)}$ , where  $f_3(x) = \left( \frac{1}{2013} \right)^{f_4(x)}$  where  $f_4(x) = \log_{2013} \log_x 2012$ , then the range of  $f_1(x)$  is -  
(A) (2, ∞) (B) (2012, ∞) (C) (0, ∞) (D) (-∞, ∞)

42. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by  $f(x) = \ln(x + \sqrt{x^2 + 1})$ , then number of solutions of  $|f^{-1}(x)| = e^{-|x|}$  is :-  
(A) 1 (B) 2 (C) 3 (D) Infinite
43.  $f(x) = [x - 1] + \{x\}^{[x]}$ ,  $x \in (1, 3)$ , then  $f^{-1}(x)$  is -  
(where  $[.]$  denotes greatest integer function and  $\{.\}$  denotes fractional part function)  
(A)  $\begin{cases} x+1 & x \in (1, 2) \\ 2+\sqrt{x-1} & x \in [2, 3) \end{cases}$  (B)  $\begin{cases} x-1 & x \in (1, 2) \\ 2-\sqrt{x-1} & x \in [2, 3) \end{cases}$   
(C)  $\begin{cases} x-1 & x \in (0, 1) \\ 2-\sqrt{x-1} & x \in [1, 2) \end{cases}$  (D)  $\begin{cases} x+1 & x \in (0, 1) \\ 2+\sqrt{x-1} & x \in [1, 2) \end{cases}$
44. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $f(x) = x^3 + ax^2 + bx - 8$ . If  $f(x) = 0$  has three real roots &  $f(x)$  is a bijective function, then  $(a + b)$  is equal to  
(A) 0 (B) 6 (C) -6 (D) 12
45. Which of the following functions is an odd function ?  
(A)  $|x - 2| + (x + 2) \operatorname{sgn}(x + 2)$  (B)  $\frac{1}{x(e^x - 1)} + \frac{1}{2x}$   
(C)  $\log(\sin x + \sqrt{1 + \sin^2 x})$  (D)  $e^{-4x}(e^{2x} - 1)^4$   
(where  $\operatorname{sgn}(x)$  denotes signum function of  $x$ )
46. Period of  $f(x) = \{x\} + \left\{x + \frac{1}{3}\right\} + \left\{x + \frac{2}{3}\right\}$  is equal to (where  $\{.\}$  denotes fractional part function)  
(A) 1 (B)  $\frac{2}{3}$  (C)  $\frac{1}{2}$  (D)  $\frac{1}{3}$
47. Let  $f(x) = 2x - \left\{\frac{x}{\pi}\right\}$  and  $g(x) = \cos x$ , where  $\{.\}$  denotes fractional part function, then period of  $g \circ f(x)$  is -  
(A)  $\frac{\pi}{2}$  (B)  $\pi$  (C)  $\frac{3\pi}{2}$  (D)  $\frac{\pi}{4}$
48. The period of the function  $\frac{\sin x + \sin 5x}{\cos x + \cos 5x}$  is -  
(A)  $\frac{\pi}{3}$  (B)  $\frac{\pi}{2}$  (C)  $\pi$  (D)  $2\pi$

49. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a real valued function such that  $f(10 + x) = f(10 - x) \forall x \in \mathbb{R}$  and  $f(20 + x) = -f(20 - x) \forall x \in \mathbb{R}$ . Then which of the following statements is true -
- (A)  $f(x)$  is odd and periodic                      (B)  $f(x)$  is odd and aperiodic  
(C)  $f(x)$  is even and periodic                      (D)  $f(x)$  is even and aperiodic

### Linked Comprehension Type

#### Paragraph for Question 50 & 51

$$\text{Let } f(x) = \begin{cases} x & ; x < 0 \\ 1-x & ; x \geq 0 \end{cases} \text{ \& } g(x) = \begin{cases} x^2 & ; x < -1 \\ 2x+3 & ; -1 \leq x \leq 1 \\ x & ; x > 1 \end{cases}$$

On the basis of above information, answer the following questions :

50. Range of  $f(x)$  is -  
(A)  $(-\infty, 1]$                       (B)  $(-\infty, \infty)$                       (C)  $(-\infty, 0]$                       (D)  $(-\infty, 2]$
51. Range of  $g(f(x))$  is -  
(A)  $(-\infty, \infty)$                       (B)  $[1, 3) \cup (3, \infty)$                       (C)  $[1, \infty)$                       (D)  $[0, \infty)$

## EXERCISE # 2

1. Find the domains of definitions of the following functions :

(Read the symbols  $[*]$  and  $\{*\}$  as greatest integers and fractional part functions respectively.)

(i)  $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$  (ii)  $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

(iii)  $f(x) = \ln(\sqrt{x^2 - 5x - 24} - x - 2)$  (iv)  $f(x) = \sqrt{\frac{1 - 5^x}{7^{-x} - 7}}$

(v)  $y = \log_{10} \sin(x - 3) + \sqrt{16 - x^2}$  (vi)  $f(x) = \log_{100x} \left( \frac{2 \log_{10} x + 1}{-x} \right)$

(vii)  $f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$  (viii)  $f(x) = \sqrt{(x^2 - 3x - 10) \ln^2(x - 3)}$

(ix)  $f(x) = \sqrt{(5x - 6 - x^2) \{ \ell n \{ x \} \}} + \sqrt{(7x - 5 - 2x^2)} + \left( \ln \left( \frac{7}{2} - x \right) \right)^{-1}$

(x)  $f(x) = \log_{\left[ \frac{x+1}{x} \right]} |x^2 - x - 6| + {}^{16-x}C_{2x-1} + {}^{20-3x}P_{2x-5}$

2. Find the domain & range of the following functions.

(i)  $y = \log_{\sqrt{5}} (\sqrt{2}(\sin x - \cos x) + 3)$  (ii)  $y = \frac{2x}{1 + x^2}$

(iii)  $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$  (iv)  $f(x) = \frac{x}{1 + |x|}$

(v)  $y = \sqrt{2 - x} + \sqrt{1 + x}$  (vi)  $f(x) = \frac{\sqrt{x + 4} - 3}{x - 5}$

3. (a) Draw graphs of the following function, where  $[ ]$  denotes the greatest integer function.

(i)  $f(x) = x + [x]$

(ii)  $y = (x)^{[x]}$  where  $x = [x] + (x)$  &  $x > 0$  &  $x \leq 3$

(iii)  $y = \operatorname{sgn}[x]$

(iv)  $y = \operatorname{sgn}(x - |x|)$

(b) Identify the pair(s) of functions which are identical ?

(where  $[x]$  denotes greatest integer and  $\{x\}$  denotes fractional part function)

(i)  $f(x) = \operatorname{sgn}(x^2 - 3x + 4)$  and  $g(x) = e^{\{[x]\}}$

(ii)  $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$  and  $g(x) = \tan x$

(iii)  $f(x) = \ln(1 + x) + \ln(1 - x)$  and  $g(x) = \ln(1 - x^2)$

(iv)  $f(x) = \frac{\cos x}{1 - \sin x}$  and  $g(x) = \frac{1 + \sin x}{\cos x}$

4. Classify the following functions  $f(x)$  defined in  $\mathbb{R} \rightarrow \mathbb{R}$  as injective, surjective, both or none.
- (a)  $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$  (b)  $f(x) = x^3 - 6x^2 + 11x - 6$
- (c)  $f(x) = (x^2 + x + 5)(x^2 + x - 3)$
5. Solve the following problems from (a) to (d) on functional equation :
- (a) The function  $f(x)$  defined on the real numbers has the property that  $f(f(x)) \cdot (1 + f(x)) = -f(x)$  for all  $x$  in the domain of  $f$ . If the number 3 is in the domain and range of  $f$ , compute the value of  $f(3)$ .
- (b) Suppose  $f$  is a real function satisfying  $f(x + f(x)) = 4f(x)$  and  $f(1) = 4$ . Find the value of  $f(21)$ .
- (c) Let  $f$  be function defined from  $\mathbb{R}^+ \rightarrow \mathbb{R}^+$ . If  $[f(xy)]^2 = x(f(y))^2$  for all positive numbers  $x$  and  $y$  and  $f(2) = 6$ , find the value of  $f(50)$ .
- (d) Let  $f$  be a function such that  $f(3) = 1$  and  $f(3x) = x + f(3x - 3)$  for all  $x$ . Then find the value of  $f(300)$ .
6. Suppose  $f(x) = \sin x$  and  $g(x) = 1 - \sqrt{x}$ . Then find the domain and range of the following functions.
- (A)  $f \circ g$  (B)  $g \circ f$  (C)  $f \circ f$  (D)  $g \circ g$
7. A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is such that  $f\left(\frac{1-x}{1+x}\right) = x$  for all  $x \neq -1$ . Prove that following.
- (a)  $f(f(x)) = x$  (b)  $f(1/x) = -f(x)$ ,  $x \neq 0$
- (c)  $f(-x - 2) = -f(x) - 2$ .
8. (a) Find the formula for the function  $f \circ g \circ h$ , given  $f(x) = \frac{x}{x+1}$ ;  $g(x) = x^{10}$  and  $h(x) = x + 3$ . Find also the domain of this function. Also compute  $(f \circ g \circ h)(-1)$ .
- (b) If  $f(x) = \max(x, 1/x)$  for  $x > 0$  where  $\max(a, b)$  denotes the greater of the two real numbers  $a$  and  $b$ . Define the function  $g(x) = f(x) f(1/x)$  and plot its graph.
9. Let  $f$  be a one-one function with domain  $\{x, y, z\}$  and range  $\{1, 2, 3\}$ . It is given that exactly one of the following statements is true and the remaining two are false.  
 $f(x) = 1$ ;  $f(y) \neq 1$ ;  $f(z) \neq 2$ . Determine  $f^{-1}(1)$
10.  $f(x) = \begin{cases} 1-x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$  and  $g(x) = \begin{cases} -x & \text{if } x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$  find  $(f \circ g)(x)$  and  $(g \circ f)(x)$ .

11. Find whether the following functions are even or odd or none :

(a)  $f(x) = \log\left(x + \sqrt{1+x^2}\right)$

(b)  $f(x) = \frac{x(a^x+1)}{a^x-1}$

(c)  $f(x) = \sin x + \cos x$

(d)  $f(x) = x \sin^2 x - x^3$

(e)  $f(x) = \sin x - \cos x$

(f)  $f(x) = \frac{(1+2^x)^2}{2^x}$

(g)  $f(x) = \frac{x}{e^x-1} + \frac{x}{2} + 1$

(h)  $f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3}$

12. (i) Write explicitly, functions of  $y$  defined by the following equations and also find the domains of definition of the given implicit functions :

(a)  $10^x + 10^y = 10$

(b)  $x + |y| = 2y$

(ii) The function  $f(x)$  is defined on the interval  $[0, 1]$ . Find the domain of definition of the functions.

(a)  $f(\sin x)$

(b)  $f(2x+3)$

(iii) Given that  $y = f(x)$  is a function whose domain is  $[4, 7]$  and range is  $[-1, 9]$ . Find the range and domain of

(a)  $g(x) = \frac{1}{3}f(x)$

(b)  $h(x) = f(x-7)$

13. Compute the inverse of the functions :

(a)  $f(x) = \ln\left(x + \sqrt{x^2+1}\right)$

(b)  $f(x) = 2^{\frac{x}{x-1}}$

(c)  $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

14. Find the inverse of  $f(x) = 2^{\log_{10} x} + 8$  and hence solve the equation  $f(x) = f^{-1}(x)$ .

15. (a) Function  $f$  &  $g$  are defined by  $f(x) = \sin x$ ,  $x \in \mathbb{R}$ ;  $g(x) = \tan x$ ,  $x \in \mathbb{R} - \left(K + \frac{1}{2}\right)\pi$

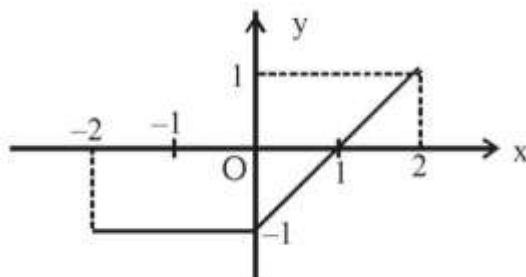
where  $K \in \mathbb{I}$ .

Find (i) periods of  $f \circ g$  &  $g \circ f$

(ii) range of the function  $f \circ g$  &  $g \circ f$

(b) Suppose that  $f$  is an even, periodic function with period 2, and that  $f(x) = x$  for all  $x$  in interval  $[0, 1]$ . Find the value of  $f(3.14)$ .

16. The graph of the function  $y = f(x)$  is as follows :



Match the function mentioned in Column-I with the respective graph given in Column-II.

**Column-I**

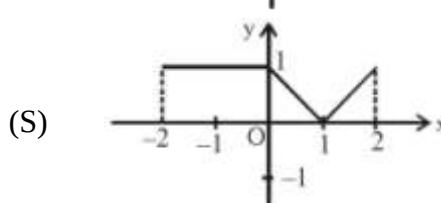
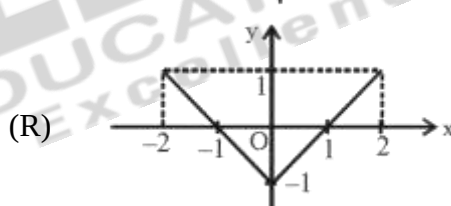
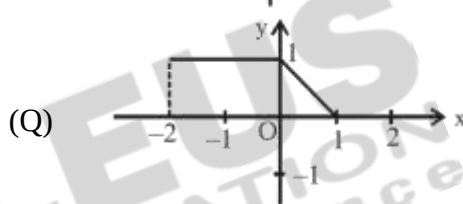
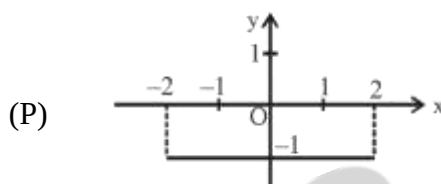
(A)  $y = |f(x)|$

(B)  $y = f(|x|)$

(C)  $y = f(-|x|)$

(D)  $y = \frac{1}{2} (|f(x)| - f(x))$

**Column-II**



17.

**Column-I**

$f(x)$

(A)  $\frac{\cos^2 x + \cos x + 2}{\cos^2 x + \cos x + 1}$

(B)  $\left| \frac{(\sqrt{\cos x} - \sqrt{\sin x})(\sqrt{\cos x} + \sqrt{\sin x})}{3(\cos x + \sin x)} \right|$

(C)  $\frac{7}{3(x^6 + 2x^4 + 3x^2 + 1)}$

(D)  $\log_8(x^2 + 2x + 2)$

**Column-II**

**Range**

(P)  $\left[0, \frac{7}{3}\right]$

(Q)  $\left[\frac{4}{3}, \frac{7}{3}\right]$

(R)  $\left[0, \frac{1}{3}\right]$

(S)  $[0, \infty)$

18. The sum of integral values of the elements in the domain of  $f(x) = \sqrt{\log_{\frac{1}{2}} |3-x|}$  is -
19. Number of integers in range of  $f(x) = x(x+2)(x+4)(x+6)+7$ ,  $x \in [-4, 2]$  is
20. The number of even integral value(s) in the range of the function  $f(x) = \frac{\tan^2 x + 8 \tan x + 15}{1 + \tan^2 x}$  is
21. If  $f(x) = a \log\left(\frac{1+x}{1-x}\right) + bx^3 + c \sin x + 5$  and  $f(\log_3 2) = 4$ , then  $f\left(\log_3\left(\frac{1}{2}\right)\right)$  is equal to
22. Let  $P(x) = x^4 + ax^3 + bx^2 + cx + d$  be a polynomial such that  $P(1) = 1$ ,  $P(2) = 8$ ,  $P(3) = 27$ ,  $P(4) = 64$  then find  $P(10)$ .
23. If  $f(x) = \begin{cases} 0 & x < 1 \\ 2x-2 & x \geq 1 \end{cases}$ ; then the number of solutions of the equation  $f(f(f(x))) = x$  is
24. Let 'f' be an even periodic function with period '4' such that  $f(x) = 2^x - 1$ ,  $0 \leq x \leq 2$ . The number of solutions of the equation  $f(x) = 1$  in  $[-10, 20]$  are
25. Let  $f(x) = \frac{2x-1}{x+3}$ . If  $f^{-1} = \frac{ax+b}{c-x}$ , then  $a + b + c$  is
26. Let  $f(x)$  be a periodic function with period 'p' satisfying  $f(x) + f(x+3) + f(x+6) + \dots + f(x+42) = \text{constant } \forall x \in \mathbb{R}$ , then sum of digits of 'p' is

### EXERCISE # 3 (JM)

- A real valued function  $f(x)$  satisfies the function equation  $f(x - y) = f(x)f(y) - f(a - x)f(a + y)$  where  $a$  is a given constant and  $f(0) = 1$ ,  $f(2a - x)$  is equal to [AIEEE-2005]

(1)  $f(1) + f(a - x)$       (2)  $f(-x)$       (3)  $-f(x)$       (4)  $f(x)$
- If  $x$  is real, the maximum value of  $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$  is - [AIEEE-2005]

(1) 41      (2) 1      (3)  $\frac{17}{7}$       (4)  $\frac{1}{4}$
- The largest interval lying in  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$  for which the function  $f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$  is defined, is [AIEEE-2007]

(1)  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$       (2)  $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$       (3)  $\left[0, \frac{\pi}{2}\right)$       (4)  $[0, \pi]$
- Let  $f : \mathbb{N} \rightarrow Y$  be a function defined as  $f(x) = 4x + 3$  where [AIEEE - 2008]  
 $Y = \{y \in \mathbb{N} : y = 4x + 3 \text{ for some } x \in \mathbb{N}\}$ . So that  $f$  is invertible and its inverse is

(1)  $g(y) = \frac{3y + 4}{3}$       (2)  $g(y) = 4 + \frac{y + 3}{4}$   
 (3)  $g(y) = \frac{y + 3}{4}$       (4)  $g(y) = \frac{y - 3}{4}$
- For real  $x$ , let  $f(x) = x^3 + 5x + 1$ , then :- [AIEEE-2009]

(1)  $f$  is one-one and onto  $\mathbb{R}$   
 (2)  $f$  is neither one-one nor onto  $\mathbb{R}$   
 (3)  $f$  is one-one but not onto  $\mathbb{R}$   
 (4)  $f$  is onto  $\mathbb{R}$  but not one-one
- The domain of the function  $f(x) = \frac{1}{\sqrt{|x| - x}}$  is :-

(1)  $(-\infty, 0)$       (2)  $(-\infty, \infty) - \{0\}$       (3)  $(-\infty, \infty)$       (4)  $(0, \infty)$

7. Let  $f$  be a function defined by  $f(x) = (x-1)^2 + 1, (x \geq 1)$  [AIEEE - 2009, 2011]  
**Statement - 1 :** The set  $\{x : f(x) = f^{-1}(x)\} = \{1, 2\}$   
**Statement - 2 :**  $f$  is bijection and  $f^{-1}(x) = 1 + \sqrt{x-1}, x \geq 1$ .  
 (1) Statement-1 is true, Statement-2 is false.  
 (2) Statement-1 is false, Statement-2 is true.  
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.  
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement-1.
8. If  $a \in \mathbb{R}$  and the equation  $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$  (where  $[x]$  denotes the greatest integer  $\leq x$ ) has no integral solution, then all possible values of  $a$  lie in the interval : [JEE(Main)-2014]  
 (1)  $(-1, 0) \cup (0, 1)$  (2)  $(1, 2)$  (3)  $(-2, -1)$  (4)  $(-\infty, -2) \cup (2, \infty)$
9. The function  $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$  defined as  $f(x) = \frac{x}{1+x^2}$ , is : [JEE(Main)-2017]  
 (1) neither injective nor surjective (2) invertible  
 (3) injective but not surjective (4) surjective but not injective
10. Let  $f : A \rightarrow B$  be a function defined as  $f(x) = \frac{x-1}{x-2}$ , where  $A = \mathbb{R} - \{2\}$  and  $B = \mathbb{R} - \{1\}$ . Then  $f$  is  
 (1) Invertible and  $f^{-1}(y) = \frac{2y-1}{y-1}$  (2) Not invertible  
 (3) Invertible and  $f^{-1}(y) = \frac{3y-1}{y-1}$  (4) Invertible and  $f^{-1}(y) = \frac{2y+1}{y-1}$  [JEE(Main) Online-2018]
11. Let  $f(x) = 2^{10} \cdot x + 1$  and  $g(x) = 3^{10} \cdot x - 1$ . If  $(f \circ g)(x) = x$ , then  $x$  is equal to  
 (1)  $\frac{3^{10}-1}{3^{10}-2^{-10}}$  (2)  $\frac{2^{10}-1}{2^{10}-3^{-10}}$  (3)  $\frac{1-3^{-10}}{2^{10}-3^{-10}}$  (4)  $\frac{1-2^{-10}}{3^{10}-2^{-10}}$  [JEE(Main) Online-2018]
12. The function  $f : \mathbb{N} \rightarrow \mathbb{N}$  defined by  $f(x) = x - 5 \left\lfloor \frac{x}{5} \right\rfloor$ , where  $\mathbb{N}$  is the set of natural numbers and  $[x]$  denotes the greatest integer less than or equal to  $x$ , is  
 (1) one-one and onto. (2) onto but not one-one.  
 (3) neither one-one nor onto. (4) one-one but not onto. [JEE(Main) Online-2018]

13. For  $x \in \mathbb{R} - \{0, 1\}$ , let  $f_1(x) = \frac{1}{x}$ ,  $f_2(x) = 1 - x$  and  $f_3(x) = \frac{1}{1-x}$  be three given functions. If a function,  $J(x)$  satisfies  $(f_2 \circ J \circ f_1)(x) = f_3(x)$  then  $J(x)$  is equal to :

- (1)  $f_3(x)$                       (2)  $f_1(x)$                       (3)  $f_2(x)$                       (4)  $\frac{1}{x} f_3(x)$

[JEE(Main) Online-2019]

14. Let  $A = \{x \in \mathbb{R} : x \text{ is not a positive integer}\}$ . Define a function  $f : A \rightarrow \mathbb{R}$  as  $f(x) = \frac{2x}{x-1}$ , then  $f$  is :

[JEE(Main) Online-2019]

- (1) neither injective nor surjective                      (2) not injective  
(3) injective but not surjective                      (4) surjective but not injective

15. Let  $\mathbb{N}$  be the set of natural numbers and two functions  $f$  and  $g$  be defined as  $f, g : \mathbb{N} \rightarrow \mathbb{N}$  such that

[JEE(Main) Online-2019]

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases} \quad \text{and } g(n) = n - (-1)^n. \text{ Then } fog \text{ is:}$$

- (1) onto but not one-one                      (2) one-one but not onto  
(3) both one-one and onto.                      (4) neither one-one nor onto.

16. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by  $f(x) = \frac{x}{1+x^2}$ ,  $x \in \mathbb{R}$ . Then the range of  $f$  is :

- (1)  $\mathbb{R} - \left[-\frac{1}{2}, \frac{1}{2}\right]$                       (2)  $\mathbb{R} - [-1, 1]$                       (3)  $\left[-\frac{1}{2}, \frac{1}{2}\right]$                       (4)  $(-1, 1) - \{0\}$

[JEE(Main) Online-2019]

17. The number of functions  $f$  from  $\{1, 2, 3, \dots, 20\}$  onto  $\{1, 2, 3, \dots, 20\}$  such that  $f(k)$  is a multiple of 3, whenever  $k$  is a multiple of 4, is:

[JEE(Main) Online-2019]

- (1)  $5^6 \times 15$                       (2)  $5! \times 6!$                       (3)  $6^5 \times (15)!$                       (4)  $(15)! \times 6!$

18. Let a function  $f : (0, \infty) \rightarrow (0, \infty)$  be defined by  $f(x) = \left|1 - \frac{1}{x}\right|$ . Then  $f$  is:

- (1) neither injective nor surjective                      (2) not injective but it is surjective  
(3) both injective as well as surjective                      (4) injective only

[JEE(Main) Online-2019]

19. If  $f(x) = \log_e \left( \frac{1-x}{1+x} \right)$ ,  $|x| < 1$ , then  $f\left(\frac{2x}{1+x^2}\right)$  is equal to

[JEE(Main) Online-2019]

- (1)  $2f(x^2)$                       (2)  $-2f(x)$                       (3)  $2f(x)$                       (4)  $(f(x))^2$

20. If  $f(1) = 1$ ,  $f'(1) = 3$ , then the derivative of  $f(f(f(x))) + (f(x))^2$  at  $x = 1$  is :  
 (1) 15 (2) 9 (3) 33 (4) 12  
**[JEE(Main) Online-2019]**
21. Let  $f(x) = a^x$  ( $a > 0$ ) be written as  $f(x) = f_1(x) + f_2(x)$ , where  $f_1(x)$  is an even function and  $f_2(x)$  is an odd function. Then  $f_1(x+y) + f_1(x-y)$  equals is :  
 (1)  $2f_1(x+y) f_2(x-y)$  (2)  $2f_1(x)f_1(y)$   
 (3)  $2f_1(x+y) f_1(x-y)$  (4)  $2f_1(x)f_2(y)$   
**[JEE(Main) Online-2019]**
22. Let  $\sum_{k=1}^{10} f(a+k) = 16(2^{10} - 1)$  where the function  $f$  satisfies  $f(x+y) = f(x) f(y)$  for all natural numbers  $x, y$  and  $f(1) = 2$ . Then the natural number 'a' is :  
 (1) 3 (2) 16 (3) 2 (4) 4  
**[JEE(Main) Online-2019]**
23. The domain of the definition of the function  $f(x) = \frac{1}{4-x^2} + \log_{10}(x^3 - x)$  is:  
 (1)  $(-1, 0) \cup (1, 2) \cup (2, \infty)$  (2)  $(-2, -1) \cup (-1, 0) \cup (2, \infty)$   
 (3)  $(1, 2) \cup (2, \infty)$  (4)  $(-1, 0) \cup (1, 2) \cup (3, \infty)$   
**[JEE(Main) Online-2019]**
24. Let  $f(x) = x^2$ ,  $x \in \mathbb{R}$ . for any  $A \subseteq \mathbb{R}$ , define  $g(A) = \{x \in \mathbb{R} ; f(x) \in A\}$ . If  $S = [0, 4]$ , then which one of the following statements is not true ?  
 (1)  $f(g(S)) \neq f(S)$  (2)  $g(f(S)) \neq S$  (3)  $g(f(S)) = g(S)$  (4)  $f(g(S)) = S$   
**[JEE(Main) Online-2019]**
25. Let  $f(x) = \log_e(\sin x)$ , ( $0 < x < \pi$ ) and  $g(x) = \sin^{-1}(e^{-x})$ , ( $x \geq 0$ ). If  $\alpha$  is a positive real number such that  $a = (f \circ g)'(\alpha)$  and  $b = (f \circ g)(\alpha)$ , then :  
 (1)  $a\alpha^2 + b\alpha - a = -2\alpha^2$  (2)  $a\alpha^2 + b\alpha + a = 0$   
 (3)  $a\alpha^2 - b\alpha - a = 1$  (4)  $a\alpha^2 - b\alpha - a = 0$   
**[JEE(Main) Online-2019]**
26. For  $x \in \left(0, \frac{3}{2}\right)$ , let  $f(x) = \sqrt{x}$ ,  $g(x) = \tan x$  and  $h(x) = \frac{1-x^2}{1+x^2}$ . If  $\phi(x) = ((h \circ f) \circ g)(x)$ , then  $\phi\left(\frac{\pi}{3}\right)$  is equal to  
**[JEE(Main) Online-2019]**  
 (1)  $\tan \frac{7\pi}{12}$  (2)  $\tan \frac{\pi}{12}$  (3)  $\tan \frac{11\pi}{12}$  (4)  $\tan \frac{5\pi}{12}$
27. For  $x \in \mathbb{R}$ , let  $[x]$  denote the greatest integer  $\leq x$ , then the sum of the series  $\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \dots + \left[-\frac{1}{3} - \frac{99}{100}\right]$  is:  
 (1) -153 (2) -133 (3) -135 (4) -131  
**[JEE(Main) Online-2019]**

## EXERCISE # 4 (JA)

1. If functions  $f(x)$  and  $g(x)$  are defined on  $\mathbb{R} \rightarrow \mathbb{R}$  such that [JEE 2005 (Scr.)]

$$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}, g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}, \text{ then } (f - g)(x) \text{ is -}$$

- (A) one-one and onto (B) neither one-one nor onto  
 (C) one-one but not onto (D) onto but not one-one

2. Let  $f(x) = x^2$  and  $g(x) = \sin x$  for all  $x \in \mathbb{R}$ . Then the set of all  $x$  satisfying  $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$ , where  $(f \circ g)(x) = f(g(x))$ , is- [JEE 2011, 3, (-1)]

- (A)  $\pm\sqrt{n\pi}$ ,  $n \in \{0, 1, 2, \dots\}$   
 (B)  $\pm\sqrt{n\pi}$ ,  $n \in \{1, 2, \dots\}$   
 (C)  $\frac{\pi}{2} + 2n\pi$ ,  $n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$   
 (D)  $2n\pi$ ,  $n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

3. The function  $f : [0, 3] \rightarrow [1, 29]$ , defined by  $f(x) = 2x^3 - 15x^2 + 36x + 1$ , is :

- (A) one-one and onto  
 (B) onto but not one-one  
 (C) one-one but not onto  
 (D) neither one-one nor onto

[JEE 2012, 3, (-1)]

4. Let  $f : (-1, 1) \rightarrow \mathbb{R}$  be such that  $f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$  for  $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ . Then the

value(s) of  $f\left(\frac{1}{3}\right)$  is (are) -

[JEE 2012, 4]

- (A)  $1 - \sqrt{\frac{3}{2}}$  (B)  $1 + \sqrt{\frac{3}{2}}$  (C)  $1 - \sqrt{\frac{2}{3}}$  (D)  $1 + \sqrt{\frac{2}{3}}$

## EXERCISE # 5

### Multiple Correct Answer Type

1. Let  $f(x) = \begin{cases} x^2 & ; 0 < x < 2 \\ 2x-3 & ; 2 \leq x < 3 \\ x+2 & ; x \geq 3 \end{cases}$  Then :-

(A)  $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\} = f\left(\frac{3}{2}\right)$

(B)  $1 + f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\} = f\left(\frac{5}{2}\right)$

(C)  $f\{f(1)\} = f(1) = 1$

(D) none of these

2. The range of the function  $f(\theta) = \sqrt{8\sin^2\theta + 4\cos^2\theta - 8\sin\theta\cos\theta}$  is -

(A)  $[\sqrt{5}-1, \sqrt{5}+1]$

(B)  $[0, \sqrt{5}+1]$

(C)  $[\sqrt{6}-\sqrt{20}, \sqrt{6}+\sqrt{20}]$

(D) none of these

3. For the function  $f(x) = |x+3| - |x+1| - |x-1| + |x-3|$ , identify correct option(s)

(A) Range of  $f(x)$  is  $(-\infty, 4]$

(B) maximum value of  $f(x)$  is 4

(C)  $f(x) = 4$  has infinite solutions

(D)  $f(x) = 0$  has infinite solutions

4. If  $\{x\} = \frac{2}{3}$  &  $\left[x + \left\{x + \left[x + \{x + \dots 100 \text{ times}\}\right]\right\}\right] = 5$ , then -

(A)  $x = \frac{14}{3}$

(B)  $[x] = 5$

(C)  $x = \frac{17}{3}$

(D)  $[x] = 4$

(where  $[.]$  &  $\{.\}$  denotes greatest integer function & fractional part function respectively)

5. The values of  $x$  in  $[-2\pi, 2\pi]$ , for which the graph of the function  $y = \sqrt{\frac{1+\sin x}{1-\sin x}} - \sec x$  and

$y = -\sqrt{\frac{1-\sin x}{1+\sin x}} + \sec x$ , coincide are

(A)  $\left[-2\pi, -\frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right]$

(B)  $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

(C)  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D)  $[-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$

6. A parabola of the form  $y = ax^2 + bx + c$  with  $a > 0$  intersects the graph of  $f(x) = \frac{1}{x^2 - 4}$ .

Number of possible distinct intersection(s) of these graph is

- (A) 0 (B) 2 (C) 3 (D) 4

7. Let  $f : \{1, 2, 3, 4, 5\} \rightarrow \{1, 2, 3, 4, 5\}$  is such that  $f(x)$  is a one-one function satisfying following condition  $f(x) = x + 1$  **if and only if**  $x$  is even (i.e.  $f(3) \neq 4$ ,  $f(4) = 5$  etc.). Then  $f^{-1}(2)$  can be-

- (A) 1 (B) 3 (C) 5 (D) 2

8. Which of the following statement(s) is(are) correct ?

- (A) If  $f$  is a one-one mapping from set  $A$  to  $A$ , then  $f$  is onto.  
(B) If  $f$  is an onto mapping from set  $A$  to  $A$ , then  $f$  is one-one  
(C) Let  $f$  and  $g$  be two functions defined from  $R \rightarrow R$  such that  $gof$  is injective, then  $f$  must be injective.  
(D) If set  $A$  contains 3 elements while set  $B$  contains 2 elements, then total number of functions from  $A$  to  $B$  is 8.

9. If  $f(x) = ax + b$  and  $f(f(f(x))) = 27x + 13$  where  $a$  and  $b$  are real numbers, then-

- (A)  $a + b = 3$  (B)  $a + b = 4$  (C)  $f'(x) = 3$  (D)  $f'(x) = -3$

10. Given  $f(1) = 2$  and  $f(n+1) = \frac{f(n)-1}{f(n)+1} \forall n \in N$ , then -

- (A)  $f(2015) = -\frac{1}{2}$  (B)  $(f(2012))^{f(2013)} = 9$   
(C)  $f(1001) = 2$  (D)  $f(2015) = -3$

11. Which of the following is/are true?

- (A)  $f(x) = e^x$  and  $g(x) = \ln x$ , then  $f(g(x)) = x$  (wherever  $f(g(x))$  is defined)  
(B)  $f(x) = \frac{2+x}{3-x}$  and  $g(x) = \frac{(3x-2)}{(x+1)}$ , then  $f(g(x)) = x$  (wherever  $f(g(x))$  is defined)  
(C)  $f(x) = 4x + 7$  and  $g(x) = \frac{(x-7)}{4}$ , then  $f(g(x)) = x$  (wherever  $f(g(x))$  is defined)  
(D)  $f(x) = x^3 + 1$  and  $g(x) = (x-1)^{1/3}$ , then  $f(g(x)) = x$  (wherever  $f(g(x))$  is defined)

12. If  $g(x) = x^2 - x + 1$  and  $f(x) = \sqrt{\frac{1}{x} - x}$ , then -

- (A) Domain of  $f(g(x))$  is  $[0, 1]$  (B) Range of  $f(g(x))$  is  $\left(0, \frac{7}{2\sqrt{3}}\right]$   
(C)  $f(g(x))$  is many-one function (D)  $f(g(x))$  is unbounded function

13. Which of the following function(s) have the same domain and range ?

- (A)  $f(x) = \sqrt{1-x^2}$  (B)  $g(x) = \frac{1}{x}$   
(C)  $h(x) = \sqrt{x}$  (D)  $l(x) = \sqrt{4-x}$

14. Let  $f(x) = x^2 + 3x + 2$ , then number of solutions to -

- (A)  $f(|x|) = 2$  is 1 (B)  $f(|x|) = 2$  is 3  
(C)  $|f(x)| = 0.125$  is 4 (D)  $|f(|x|)| = 0.125$  is 8

15. Which of the following pair(s) of function have same graphs?

- (A)  $f(x) = \frac{\sec x}{\cos x} - \frac{\tan x}{\cot x}$ ,  $g(x) = \frac{\cos x}{\sec x} + \frac{\sin x}{\operatorname{cosec} x}$   
(B)  $f(x) = \operatorname{sgn}(x^2 - 4x + 5)$ ,  $g(x) = \operatorname{sgn}\left(\cos^2 x + \sin^2\left(x + \frac{\pi}{3}\right)\right)$

where  $\operatorname{sgn}$  denotes signum function.

- (C)  $f(x) = e^{\ln(x^2 + 3x + 3)}$ ,  $g(x) = x^2 + 3x + 3$   
(D)  $f(x) = \frac{\sin x}{\sec x} + \frac{\cos x}{\operatorname{cosec} x}$ ,  $g(x) = \frac{2\cos^2 x}{\cot x}$

16. If a function is defined by an implicit equation  $2^{|x|+|y|} + 2^{|x|-|y|} = 2$ , then -

- (A) Domain of function is singleton  
(B) Range of function is singleton  
(C) Graph of the function intersects the line  $y = x$   
(D) Maximum value of function is 2

17. For each real  $x$ , let  $f(x) = \max\{x, x^2, x^3, x^4\}$ , then  $f(x)$  is -

- (A)  $x^4$  for  $x \leq -1$  (B)  $x^2$  for  $-1 < x \leq 0$   
(C)  $f\left(\frac{1}{2}\right) = \frac{1}{2}$  (D)  $f\left(\frac{1}{2}\right) = \frac{1}{4}$

18. Let  $f(x) = \sin^6 x + \cos^6 x$ , then -  
 (A)  $f(x) \in [0, 1] \forall x \in \mathbb{R}$   
 (B)  $f(x) = 0$  has no solution  
 (C)  $f(x) \in \left[\frac{1}{4}, 1\right] \forall x \in \mathbb{R}$   
 (D)  $f(x)$  is an injective function
19. Let  $f(x) = \begin{cases} x^2 - 3x + 4 & ; x < 3 \\ x + 7 & ; x \geq 3 \end{cases}$  and  $g(x) = \begin{cases} x + 6 & ; x < 4 \\ x^2 + x + 2 & ; x \geq 4 \end{cases}$  then which of the following is/are true -  
 (A)  $(f + g)(1) = 9$   
 (B)  $(f - g)(3.5) = 1$   
 (C)  $(f \cdot g)(0) = 24$   
 (D)  $\left(\frac{f}{g}\right)(5) = \frac{8}{3}$

### Linked Comprehension Type

#### Paragraph for Question 20 to 22

$$f(x) = \frac{x^2 + 1}{ax} \quad (a \neq 0); \quad g(x) = 3 \sec x; \quad h(x) = \frac{x + 3}{x - 4}.$$

On the basis of above information, answer the following questions :

20. If range of  $f(x)$  and  $g(x)$  are equal sets then 'a' is equal to -  
 (A) 3  
 (B)  $-2/3$   
 (C)  $3/2$   
 (D)  $-3/2$
21.  $f(x)$  is one-one if-  
 (A)  $x \in (0, \infty)$   
 (B)  $x \in (-\infty, 0)$   
 (C)  $x \in (1, \infty)$   
 (D)  $x \in (-\infty, 1) - \{0\}$
22. Which the following is always false ?  
 (A)  $h(x)$  is one-one  
 (B)  $f(x)$  is one-one if  $x > 10$   
 (C)  $g(x)$  is many-one if  $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 3\right)$   
 (D) The values of k for which  $f(x) = k$  has exactly one solution is  $k = 2$  or  $k = -2$

**Matrix Match Type**

23. **Column-I**  
**Number of integers in**
- (A) Domain of  $f(x) = \ln\{x\}$  (P) 0
- (B) Domain of  $f(x) = \sqrt{\sec(\sin x)} + \sqrt{\left[x + \frac{1}{x}\right]} + \sqrt{10 - [x]^2}$  (Q) 2
- (C) Range of  $f(x) = x^2 - 2x + 2, x \in [0, 2]$  (R) 3
- (D) Range of  $f(x) = \sqrt{25 - [x]^2}$  (S) less than 3  
(T) more than 3
- (where  $[.]$  and  $\{.\}$  denote greatest integer function and fractional part function respectively)

24. Match the function mentioned in column-I with the respective classification given in column-II.  
(where  $[.]$  and  $\{.\}$  denote greatest integer function and fractional part function respectively)

- | <b>Column-I</b>                                                                      | <b>Column-II</b>              |
|--------------------------------------------------------------------------------------|-------------------------------|
| (A) $f : \mathbb{R} \rightarrow \mathbb{R}^+ f(x) = (e^{[x]})(e^{\{x\}})$            | (P) one-one                   |
| (B) $f : (-\infty, -2) \cup (0, \infty) \rightarrow \mathbb{R} f(x) = \ln(x^2 + 2x)$ | (Q) many-one                  |
| (C) $f : [-2, 2] \rightarrow [-1, 1] f(x) = \sin x$                                  | (R) onto                      |
| (D) $f : \mathbb{R} \rightarrow \mathbb{R} f(x) = x^3 - 3x^2 + 3x - 7$               | (S) periodic<br>(T) aperiodic |

25. Match the functions given in column-I correctly with mappings given in column-II.

- | <b>Column-I</b>                                                                                    | <b>Column-II</b>                                    |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| (A) $f : \left[-\frac{1}{2}, \frac{1}{2}\right] \rightarrow \left[\frac{4}{7}, \frac{4}{3}\right]$ | (P) Injective mapping                               |
| $f(x) = \frac{1}{x^2 + x + 1}$                                                                     | (Q) Non-injective mapping                           |
| (B) $f : [-2, 2] \rightarrow [-1, 1]$<br>$f(x) = \sin x$                                           | (R) Surjective mapping                              |
| (C) $f : \mathbb{R} - \mathbb{I} \rightarrow \mathbb{R}$<br>$f(x) = \ln\{x\},$                     | (S) Non-surjective mapping<br>(T) Bijective mapping |
| (where $\{.\}$ represents fractional part function)                                                |                                                     |
| (D) $f : (-\infty, 0] \rightarrow [1, \infty), f(x) = (1 + \sqrt{-x}) + (\sqrt{-x} - x)$           |                                                     |

## EXERCISE # 6

1. Find the domains of definitions of the following functions :  
 (Read the symbols  $[*]$  as greatest integers function)
 

(i)  $f(x) = \sqrt{\log_x (\cos 2\pi x)}$

(ii)  $f(x) = \frac{[x]}{2x - [x]}$
  
2. (a) Let  $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$  be a polynomial such that  $P(1) = 1$ ;  $P(2) = 2$ ;  $P(3) = 3$ ;  $P(4) = 4$ ;  $P(5) = 5$  and  $P(6) = 6$ , then find the value of  $P(7)$ .  
 (b) Let  $a$  and  $b$  be real numbers and let  $f(x) = a \sin x + b\sqrt[3]{x} + 4$ ,  $\forall x \in \mathbb{R}$ . If  $f(\log_{10}(\log_3 10)) = 5$  then find the value of  $f(\log_{10}(\log_{10} 3))$ .
  
3. A function  $f$ , defined for all  $x, y \in \mathbb{R}$  is such that  $f(1) = 2$ ;  $f(2) = 8$  &  $f(x+y) - kxy = f(x) + 2y^2$ , where  $k$  is some constant. Find  $f(x)$  & show that :  

$$f(x+y)f\left(\frac{1}{x+y}\right) = k \text{ for } x+y \neq 0.$$
  
4. If  $f(x) = -1 + |x - 2|$ ,  $0 \leq x \leq 4$   
 $g(x) = 2 - |x|$ ,  $-1 \leq x \leq 3$   
 Then find  $f \circ g(x)$  &  $g \circ f(x)$ . Draw rough sketch of the graphs of  $f \circ g(x)$  &  $g \circ f(x)$ .
  
5. Let  $\{x\}$  &  $[x]$  denote the fractional and integral part of a real number  $x$  respectively. Solve  $4\{x\} = x + [x]$ .
  
6. Let  $f(x) = \frac{9^x}{9^x + 3}$  then find the value of the sum  $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$
  
7. Let  $f(x) = (x+1)(x+2)(x+3)(x+4) + 5$  where  $x \in [-6, 6]$ . If the range of the function is  $[a, b]$  where  $a, b \in \mathbb{N}$  then find the value of  $(a+b)$ .
  
8.  $f(x)$  and  $g(x)$  are linear function such that for all  $x$ ,  $f(g(x))$  and  $g(f(x))$  are Identity functions. If  $f(0) = 4$  and  $g(5) = 17$ , compute  $f(2006)$ .

**Comprehension for Q.9 to Q.11**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a function satisfying  $f(2 - x) = f(2 + x)$  and  $f(20 - x) = f(x)$ ,  $\forall x \in \mathbb{R}$ .

**On the basis of above information, answer the following questions :**

9. If  $f(0) = 5$ , then minimum possible number of values of  $x$  satisfying  $f(x) = 5$ , for  $x \in [0, 170]$  is-
- (A) 21                      (B) 12                      (C) 11                      (D) 22
10. Graph of  $y = f(x)$  is -
- (A) symmetrical about  $x = 18$                       (B) symmetrical about  $x = 5$   
(C) symmetrical about  $x = 8$                       (D) symmetrical about  $x = 20$
11. If  $f(2) \neq f(6)$ , then
- (A) fundamental period of  $f(x)$  is 1                      (B) fundamental period of  $f(x)$  may be 1  
(C) period of  $f(x)$  can't be 1                      (D) fundamental period of  $f(x)$  is 8

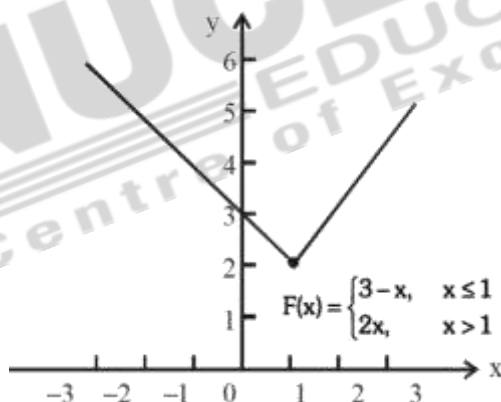
**Subjective Type**

12. The function  $f(x)$  has the property that for each real number  $x$  in its domain,  $1/x$  is also in its domain and  $f(x) + f(1/x) = x$ . Find the largest set of real numbers that can be in the domain of  $f(x)$ ?

## FUNCTIONS

### EXERCISE # 1

1. (i)  $-\infty < x \leq 0$  (ii)  $x \in \mathbb{R}$  (iii)  $x \in \mathbb{R} - \{-1, 0, 1\}$   
 (iv)  $-\infty < x < 0$  &  $4 < x < \infty$  (v)  $-\infty < x \leq 1$  and  $3 \leq x < \infty$  (vi)  $-\infty < x < 1$  and  $2 < x < \infty$   
 (vii)  $-1 \leq x \leq 1$  (viii)  $0 < x < 1$  and  $1 < x < \infty$ . (ix)  $-2 \leq x < 0$  and  $0 < x < 1$   
 (x)  $\frac{3}{2} < x < 2$  and  $2 < x < \infty$ . (xi)  $-1 < x < 0$  and  $1 < x < 2$ ;  $2 < x < \infty$   
 (xii)  $2k\pi < x < (2k + 1)\pi$ , where  $k$  is an integer. (xiii)  $4 \leq x \leq 6$  (xiv)  $2 < x < 3$
2. (i)  $\mathbb{R} - \{1\}$  (ii)  $\mathbb{R} - \{0\}$  (iii)  $(0, 4/3]$  (iv)  $[1/3, 3]$  (v)  $[1, \infty)$  (vi)  $\mathbb{R}$   
 (vii)  $[0, \infty)$  (viii)  $[3, 7]$  (ix)  $[-1, 1]$  (x)  $[2, \infty)$  (xi)  $[1/3, 1]$  (xii)  $\left[\frac{1}{11}, \frac{1}{5}\right]$
3. (a)  $-2$ , (b)  $-3, 1$ , (c)  $[-3, 3], [-2, 3]$ , (d)  $(-1, 3]$ , (e)  $B$ , (f)  $A, D$
- 4.



5. (a)  $y = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$  (b)  $y = \begin{cases} 2, & 0 \leq x < 1 \text{ and } 2 \leq x < 3 \\ 0, & 1 \leq x < 2 \text{ and } 3 \leq x \leq 4 \end{cases}$
6. (a)  $-2, 4$ , (b)  $-2, 2$ , (c)  $-3, 4$ , (d)  $(0, 4]$  (e)  $[-4, 4], [-2, 3]$   
 (f)  $[-4, 3], [0.5, 4]$
7. (a)  $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$  (b)  $[-2, -1) \cup [1, 3]$  (c)  $[-3, -2] \cup [-1, 1]$  (d)  $\left(\frac{5}{3}, 3\right) - \{2\}$

|     |   |     |   |     |   |     |   |     |   |     |   |     |   |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 8.  | C | 9.  | D | 10. | C | 11. | A | 12. | C | 13. | B | 14. | C |
| 15. | C | 16. | D | 17. | B | 18. | C | 19. | A | 20. | C | 21. | D |
| 22. | B | 23. | D | 24. | B | 25. | B | 26. | C | 27. | D | 28. | D |
| 29. | D | 30. | C | 31. | D | 32. | C | 33. | B | 34. | D | 35. | C |
| 36. | B | 37. | A | 38. | B | 39. | A | 40. | D | 41. | A | 42. | B |
| 43. | D | 44. | B | 45. | C | 46. | D | 47. | B | 48. | C | 49. | A |
| 50. | A | 51. | C |     |   |     |   |     |   |     |   |     |   |

### EXERCISE # 2

1. (i)  $\left[-\frac{5\pi}{4}, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$  (ii)  $\left(-4, -\frac{1}{2}\right) \cup (2, \infty)$
- (iii)  $(-\infty, -3]$  (iv)  $(-\infty, -1) \cup [0, \infty)$
- (v)  $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \leq 4)$  (vi)  $\left(0, \frac{1}{100}\right) \cup \left(\frac{1}{100}, \frac{1}{\sqrt{10}}\right)$
- (vii)  $(-3, -1] \cup \{0\} \cup [1, 3)$  (viii)  $\{4\} \cup [5, \infty)$
- (ix)  $(1, 2) \cup (2, 5/2)$  (x)  $x \in \{4, 5\}$
2. (i)  $D : x \in \mathbb{R} \quad R : [0, 2]$
- (ii)  $D = \mathbb{R}; \text{ range } [-1, 1]$
- (iii)  $D : \{x \mid x \in \mathbb{R}; x \neq -3; x \neq 2\} \quad R : \{f(x) \mid f(x) \in \mathbb{R}, f(x) \neq 1/5; f(x) \neq 1\}$
- (vi)  $D : \mathbb{R}, R : (-1, 1)$
- (v)  $D : -1 \leq x \leq 2 \quad R : [\sqrt{3}, \sqrt{6}]$
- (iv)  $D : [-4, \infty) - \{5\}; R : \left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$

3. (b) (i), (iii) are identical
4. (a) neither surjective nor injective  
(b) surjective but not injective  
(c) neither injective nor surjective
5. (a)  $-3/4$ ; (b) 64; (c) 30, (d) 5050
6. (a) domain is  $x \geq 0$ ; range  $[-1, 1]$   
(b) domain  $2k\pi \leq x \leq 2k\pi + \pi$ ; range  $[0, 1]$   
(c) domain  $x \in \mathbb{R}$ ; range  $[-\sin 1, \sin 1]$   
(d) domain is  $0 \leq x \leq 1$ ; range is  $[0, 1]$
8. (a)  $\frac{(x+3)^{10}}{(x+3)^{10}+1}$  domain is  $\mathbb{R}$ ,  $\frac{1024}{1025}$   
(b)  $g(x) = \begin{cases} \frac{1}{x^2} & \text{if } 0 < x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$
9.  $f^{-1}(1) = y$
10.  $(g \circ f)(x) = \begin{cases} x & \text{if } x \leq 0 \\ -x^2 & \text{if } 0 < x < 1 \\ 1-x^2 & \text{if } x \geq 1 \end{cases}; (f \circ g)(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1+x & \text{if } 0 \leq x < 1 \\ x & \text{if } x \geq 1 \end{cases}$
11. (a) odd, (b) even, (c) neither odd nor even, (d) odd,  
(e) neither odd nor even, (f) even, (g) even, (h) even
12. (i) (a)  $y = \log(10 - 10^x)$ ,  $-\infty < x < 1$   
(b)  $y = x/3$  when  $-\infty < x < 0$  &  $y = x$  when  $0 \leq x < +\infty$   
(ii) (a)  $2K\pi \leq x \leq 2K\pi + \pi$  where  $K \in \mathbb{I}$  (b)  $[-3/2, -1]$   
(iii) (a) Range :  $[-1/3, 3]$ , Domain =  $[4, 7]$ ; (b) Range  $[-1, 9]$  and domain  $[11, 14]$

13. (a)  $\frac{e^x - e^{-x}}{2}$ ; (b)  $\frac{\log_2 x}{\log_2 x - 1}$ ; (c)  $\frac{1}{2} \log \frac{1+x}{1-x}$

14.  $x = 10$ ;  $f^{-1}(x) = 10^{\log_2(x-8)}$

15. (a) (i) Period of  $f \circ g$  is  $\pi$ , period of  $g \circ f$  is  $2\pi$   
(ii) range of  $f \circ g$  is  $[-1, 1]$ , range of  $g \circ f$  is  $[-\tan 1, \tan 1]$   
(b) 0.86

16. (A) S; (B) R; (C) P; (D) Q

17. (A)  $\rightarrow$  (Q); (B)  $\rightarrow$  (R); (C)  $\rightarrow$  (P); (D)  $\rightarrow$  (S)

18. 6      19. 401      20. 9      21. 6      22. 4024      23. 2      24. 15

25. 6      26. 9

### EXERCISE # 3 (JM)

1. 3      2. 1      3. 3      4. 4      5. 1      6. 1      7. 3  
8. 1      9. 4      10. 1      11. 4      12. 3      13. 1      14. 3  
15. 1      16. 3      17. 4      18. Bonus      19. 3      20. 3      21. 2  
22. 1      23. 1      24. 3      25. 3      26. 3      27. 2

### EXERCISE # 4 (JA)

1. A      2. A      3. B      4. (zero marks to all)

### EXERCISE # 5

1. A,B,C      2. A,C      3. B,C,D      4. A,D      5. A,C  
6. B,C,D      7. B,C      8. C,D      9. B,C      10. A,B,C  
11. A,B,C,D      12. A,C      13. B,C      14. A,C  
15. A,B,C,D      16. A,B,C      17. A,B,C      18. B,C      19. A,B,C  
20. B      21. C      22. C  
23. (A) $\rightarrow$ (P,S); (B) $\rightarrow$ (R); (C)  $\rightarrow$  (Q,S); (D)  $\rightarrow$  (T)  
24. (A) $\rightarrow$ (P,R,T); (B)  $\rightarrow$  (Q,R,T); (C)  $\rightarrow$  (Q,R,T); (D)  $\rightarrow$  (P,R,T)  
25. (A) $\rightarrow$  (P,R,T); (B) $\rightarrow$  (Q,R); (C) $\rightarrow$  (Q,S); (D) $\rightarrow$  (P,R,T)

## EXERCISE # 6

1. (i)  $(0, 1/4) \cup (3/4, 1) \cup \{x : x \in \mathbb{N}, x \geq 2\}$  (ii)  $\mathbb{R} - \left\{-\frac{1}{2}, 0\right\}$

2. (a) 727, (b) 3 3.  $f(x) = 2x^2$

4.  $f \circ g(x) = \begin{cases} -(1+x), & -1 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}; f \circ f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 3-x, & 1 \leq x \leq 2 \\ x-1, & 2 < x \leq 3 \\ 5-x, & 3 < x \leq 4 \end{cases}$

$f \circ f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 4-x, & 3 \leq x \leq 4 \end{cases}; g \circ g(x) = \begin{cases} -x, & -1 \leq x \leq 0 \\ x, & 0 < x \leq 2 \\ 4-x, & 2 < x \leq 3 \end{cases}$

5.  $x = 0$  or  $5/3$  6. 1002.5 7. 5049 8. 122

9. D 10. A 11. C 12.  $\{-1, 1\}$